BEYOND EVALUATION USE

An Empirical Study of Involvement and Use/Influence in Multi-Site Evaluations

Report 3:
A Case Study of Involvement, Use, and Influence within the Collaboratives for Excellence in Teacher Preparation (CETP) Core Evaluation

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Executive Summary

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean A. King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the science, technology, engineering, and mathematics (STEM) education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of STEM education and evaluation were secondary audiences, not the directly intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

Data were collected through an online survey distributed to evaluation project leaders[i] and evaluators, interviews with project leaders and evaluators, an interview with one of the NSF program officers responsible for the program at the time, an on line survey of NSF education project PIs and questions of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the evaluation projects and through reflective descriptions provided by the evaluation projects principal investigators.

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the CETP program evaluation project. The national CETP evaluation was instituted after many of the individual CETP projects had been in operation for some time. Through meetings and other types of communication the CETP evaluation team attempted to involve the existing projects in the design of the evaluation, the development of evaluation questions, the construction of data collection instruments, and data collection and analyses.

This case study showed a modest relationship between CETP project involvement in the CETP evaluation and influence of the evaluation's processes and products on the projects. It was clear that involvement was not the only variable affecting influence. Furthermore, the effect of involvement was confounded by a wide range of perceptions about what constitutes involvement. Despite similar opportunities to participate in CETP evaluation activities, interviewees from some projects felt that they were heavily involved while others felt they were less involved and the activities cited as contributing to the involvement varied in both breadth and depth. For some, in depth involvement in one activity was seen as constituting substantial involvement while for others participation in several different activities was necessary. It did appear from the surveys and interviews with project personnel, however, that

perceptions of levels of involvement were positively linked to perceptions of use. Other variables affecting use included where the project was in its life cycle, the perceived relevance of the CETP evaluation to the local project and interactions with NSF.

The CETP evaluation had only a very small discernable effect on the fields of STEM education and evaluation. Although responses were mixed, overall through the eyes of the project personnel, the CETP evaluation was viewed as having a small effect on the fields of science education and evaluation and that this effect was probably diffused and not really attributable directly to the CETP evaluation. The citation analyses of use of the products of the CETP evaluation showed that there was some use particularly of the instruments by people in CETP projects and elsewhere. The survey of NSF PIs revealed that the respondents had little knowledge of evaluations in general and practically no knowledge of the CETP evaluation.

In summary, the CETP evaluation had a modest impact on the CETP projects that was enhanced by the involvement of the projects in the evaluation and only a very small effect on the fields of STEM education and evaluation.

Introduction

In November 2004, the National Science Foundation (NSF) awarded Drs. Frances Lawrenz and Jean King of the University of Minnesota a grant ("Beyond Evaluation Use") to study the relationship between involvement and use in multi-site evaluations. The overall question guiding this research is: What is the relationship between involvement in evaluation and use of the evaluation processes and outcomes in multi-site evaluations? The grant activities were aimed at describing and measuring how four NSF-funded program-level evaluations involved projects in the multi-site evaluation and, in turn, how this level of involvement affected the ways in which individuals, the projects, and the science, technology, engineering, and mathematics (STEM) education and evaluation communities used the process or results. NSF was the primary "client" for the evaluation. Therefore, the individual projects and the fields of STEM education and evaluation were secondary audiences, not directly the intended users.

Evaluation projects related to the following four NSF programs were included in the study: Advanced Technological Education (ATE), Collaboratives for Excellence in Teacher Preparation (CETP), Local Systemic Change through Teacher Enhancement (LSC), and Mathematics and Science Partnerships – Research, Evaluation and Technical Assistance (MSP-RETA).

This case study examines the relationship between involvement and use in the program-level evaluation of one of these programs – the CETP program evaluation project. The case study has three sections. First, it describes the history of the CETP core evaluation and the context within which the evaluation was conducted. Second, it examines the relationship between project-level involvement in the evaluation and project use of the evaluation processes and outcomes. And finally, it describes the extent to which the CETP core evaluation influenced the fields of STEM education and evaluation.

Methods

Data were collected through an online survey distributed to CETP project leaders[ii] and evaluators, interviews with CETP project leaders and evaluators, an interview with one of the NSF program officers responsible for the CETP program at the time, an online survey of NSF education project Principal Investigators (PIs), and a short survey of journal editors. Additional data were collected through review of archival documents, publications, and reports produced by the CETP program and core evaluation and through reflective descriptions provided by the CETP core evaluation PI.

CETP Project Leader and Evaluator Survey

The research team acquired names and email addresses for the PIs, coco-PIs and evaluators of CETP projects. The team compiled lists from project and meeting rosters provided by the program evaluation personnel. When email addresses were not provided for individuals on the list, a member of the team searched the Internet for an email address. In all, the research team acquired names and email addresses for a total of 99 CETP representatives.

The CETP evaluators and project leaders received an initial invitation email with a description of the survey, an explanation of its use, the Human Subjects approval number, and phone numbers to call with questions. The email also contained a link to the online survey with a login and password. Two and four weeks following the first email, the team sent reminders to nonrespondents. The data were collected from August 10, 2006 through January 17, 2007.

Complete details of the survey's development, administration, and validation are presented in "Beyond Evaluation Use Report 1: Findings from the NSF Follow-up Survey." Survey findings presented in this report are taken from 57 closed-ended items and four open-ended questions related to the involvement in and use of the CETP program evaluation.

A total of 54 surveys were completed from CETP project representatives for an overall response rate of 55%. Of the total respondents, 19 were CETP evaluators, 33 were CETP project leaders, and two were CETP project administrators.

Responses to open-ended items were reviewed by team members. When appropriate, responses may be used to support or exemplify findings from the closed-ended survey items or the project interview data, which is described in the next section of this report. The following table shows the number and percentage of respondents who provided comments to each open-ended question or statement.

Table 1: Responses to open-ended items

Type of Comment	Comment[iii]	No Comment	Percent Respondents with Comments
Example of a negative influence of the program evaluation	9	45	16.7%
Example of factors that enabled use	32	22	59.3%
Example of factors that limited use	20	34	37%
Example of inappropriate use	1	53	1.9%
General comments about the program evaluation	22	29	40.7%

Nonrespondent Study. To check for nonresponse bias, the team conducted a nonrespondent follow-up survey after the online survey of the PIs and evaluators. We developed a three-question nonrespondent survey to address the topics of involvement and use, as well as to gather some information about what kept individuals from responding to the initial survey and reminder emails. The nonrespondent survey items were:

1. How involved were you in the CETP Program Evaluation?
2. How much impact did the CETP Program Evaluation have on you?
3. What was the main thing that kept you from responding to our initial survey request?

A random sample of nonrespondents to the survey was drawn using SPSS. The team sent each person an individually addressed email with a request for input on the three items. Two reminder emails were sent to each of the respondents who did not respond to the initial request. Then, if email contact did not solicit a response, a team member contacted the individual by telephone. Fifty-seven percent of the nonrespondents (17/30) completed the nonrespondent survey.

The results of the analysis comparing the CETP respondents and nonrespondents showed no significant difference in the overall level of involvement reported (p=.066) but did show a difference in the overall level of impact (p=.013). Means of the survey's involvement and use items were used to measure level of involvement and impact for respondents for comparison to the nonrespondents (see Table 2). The process of distilling the larger survey's multiple items related to use into a single question resulted in the choice of the word "impact" as a global concept to encompass questions about use of the process, instruments, and results of the evaluation. Therefore, the difference between respondents and nonrespondents may indicate the possibility of upward bias in the estimates of overall use for CETP survey respondents or it may also be an artifact of the difference between use, influence and overall impact. Regardless, caution should be used when drawing conclusions about the true levels of impact on the population of CETP evaluators and leaders because of this observed non-response bias.

Table 2. Results of nonrespondent study

	Involvement Mean			**Use/Impact Mean**		
CETP	Respondents	Non-Respondents	p-value	Respondents	Non-Respondents	p-value
	2.38	1.82	0.066	2.52	1.76	**0.013**

Interviews

Research team members conducted in-depth interviews with a purposively selected sample of four project leaders and evaluators who completed the online survey (the Project Leader and Evaluator survey) and said they were willing to be interviewed. Survey respondents' responses to involvement and use questions were used to identify four key types of respondents: 1) high involvement, high use; 2) high involvement, low use; 3) low involvement, high use; and 4) low involvement, low use.

These categories were determined based on relative reported levels of involvement and use, with a mean response of 2.5 on the use and involvement questions as the approximate cut-off point between high and low levels. Interview participants were selected based not only on their responses to the survey but also to balance project leaders and evaluators and ensure diversity in the number of projects sampled.

Interviewees were contacted by email and invited to participate. The interviews were conducted in two stages. In the first stage, four individuals were initially selected as they

represented a range of use and involvement levels, although one additional individual also joined in one of the interviews bringing the total number to five. Three of the interviews were conducted with a single respondent and one or two University of Minnesota project team members. One of the interviews was conducted with two project representatives and two University of Minnesota project team members. The interviews followed a standardized, semi-structured protocol with flexibility in probing for additional details and description as necessary. The length of the interviews ranged from 30 to 45 minutes. All interviews were taped and transcribed.

In the second stage, individuals who reported high use of the evaluation were invited to be interviewed. In the end, three interviews with "high users" were conducted. Again the interviewees were tape recorded and transcribed, although a technical error prevented the tape from being usable for one of the interviewees and therefore only the interviewer's notes were used. Table 3 provides a description of the eight individuals interviewed in this study in terms of their levels of use and involvement and their status as evaluators, PIs, or other staff members.

Table 3. Description of interviewees

	Low Use	**High Use**
Low Involvement	Interviewee 1 (Project Director)	Interviewee 2 (PI)
High Involvement	Interviewee 5 (PI)	Interviewee 3 (Evaluator) Interviewee 4 (Evaluator) Interviewee 6 (PI) Interviewee 7 (Evaluator) Interviewee 8 (Evaluator)

In addition, Frances Lawrenz, CETP core evaluation principal investigator and principal investigator for the University of Minnesota project, conducted an interview with one of the NSF program officers responsible for the CETP program. The University of Minnesota team felt that Dr. Lawrenz was the best person to conduct the interview due to her long-time relationship with NSF. To prevent possible biases from affecting the data, the interview, lasting approximately 25 minutes, was taped, transcribed, and analyzed both by Dr. Lawrenz and one other University of Minnesota team member.

Citation Analysis

The citation analysis used bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work. Web of Science, Google Scholar, and Google were used to collect citation information from lists of evaluation products produced by the CETP program evaluation (see Appendices A and B for lists of the evaluation products and citing works).

The list of the CETP evaluation products was initially compiled by a University of Minnesota team member. The list was then forwarded to Horizon Research Inc. (HRI) staff members who reviewed the list and made changes. The CETP core evaluation PI was then provided with the opportunity to review the final list. Products were defined as any publications, presentations, instruments, or other related materials that were produced as part of the NSF-funded evaluation project.

The Web of Science, Google Scholar, and Google search engines were used to collect citation information from the evaluation product list. Searching was completed in late 2007 during the following periods: Web of Science October 23-25, Google Scholar October 25-November 13, and Google November 13 – December 2.

Data were collected manually by entering full reference information about the citations into an Excel spreadsheet which was later checked for accuracy and compliance with APA style and converted into an SPSS data file. Complete details of the rationale, development, administration, and validation of the citation analysis are presented in "Beyond Evaluation Use: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields, Report 6."

NSF PI Survey

As a measure of influence on the field, the research team administered an online survey to a sample of 350 individuals randomly selected from the database of over 2,000 currently-funded NSF Education and Human Development Directorate Principal Investigators (PIs). The survey elicited responses about PIs' perceptions of evaluations, people, and publications that might have been influential on the fields of STEM evaluation and education. The overall response rate was 54.7%. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Journal Editor Inquiry

A very brief email/telephone survey of editors and coeditors of six journals in the evaluation and STEM education fields was also conducted to determine if the editors were aware of any articles that had been published about influential STEM evaluations. Responses were obtained from three of six editors/coeditors. Complete details of the survey's development, administration, and analysis are presented in "Beyond Evaluation Use Report 7: Findings from the Survey of Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education Field."

Archival Review

The archival documents reviewed for CETP included reports, published instruments, and articles. We used NSF publications summarizing program activities and identifying grantees, project reports published by individual projects and centers, core evaluation instruments and handbooks, and articles published in scholarly journals based on the core evaluation and on individual project-level activity.

The collection of archival documents was primarily conducted using electronic databases and search engines. The NSF website provided information on the CETP program as a whole, as well as on individual projects. In addition, the CETP core evaluation website made the instruments and handbooks available. Finally, we conducted searches using Google, several electronic indexes including NSF's award search function, Academic Search-Premier, WilsonWeb (the H.W. Wilson Information Retrieval System for the World Wide Web), ERIC (the Education Resources Information Center), and the NSF-sponsored OERL (the Online Evaluation Resource Library). Complete details of the document collection and analysis are presented in "Beyond Evaluation Use Report 6: A Citation Analysis of the Influence of the ATE, CETP, LSC, and MSP-RETA Evaluations on the STEM Education and Evaluation Fields.

CETP Core Evaluation Background and Context

In the early 1990s, the National Science Foundation responded to the national need for more well-qualified math and science teachers by creating the Collaboratives for Excellence in Teacher Preparation (CETP) program. CETP sought to improve teacher preparation by enhancing science, mathematics, and educational methods courses at the college level. The program also sought to increase collaboration within and between K-12 and higher education institutions to foster the development of well-trained teachers competent in science and math.[iv] As the name indicates, collaboration was the hallmark of the CETP program. To be eligible, projects had to demonstrate institutional support and collaboration among two-year, four-year, and research institutions, school districts, the business community, and state departments of education.

From 1993 – 2000, NSF funded 19 CETPs, each involving 3-15 higher education institutions (including colleges, universities, and community colleges) and several school districts. Individual projects could apply for up to $1 million per year for three to five years (system-wide-focus awards.)[v] After the initial awards, CETPs could apply for three additional years of funding, in part, to cover the costs of evaluation activities (follow-up awards.)

CETP Program

The National Science Foundation's Collaboratives for Excellence in Teacher Preparation (CETP) program was a response to the national need to produce and retain increasing numbers of well-qualified teachers of mathematics and science. CETP promoted improving the science, mathematics, and technology preparation of future K-12 teachers, by enhancing college science, mathematics, and educational methods and increasing collaboration within and between K-12 and higher education institutions to foster the development of teachers with subject matter competence and the ability to teach effectively in these disciplines.

CETP Program Facts

Period: 1993-2000

Purpose: To achieve significant and systemic improvement in the science, technology, engineering, and mathematics (STEM) preparation of prospective preK-12 teachers.

Primary Target: Prospective preK-12 teachers.

Sites Funded: 19 sites each involving 3-15 higher education institutions with awards of up to $1,000,000 per year for 3-5 years

NSF required individual CETPs to focus on monitoring and evaluation to meet the federal requirements of the Government Performance and Results Act (GPRA). However, even as the CETP projects were conducting largely formative evaluation activities, GPRA was resulting in a shift in emphasis toward measurable outcomes and summative evaluations.

CETP teacher preparation programs required full institutional support and the concerted effort of many stakeholders, including faculty and administration from two-year, four-year, and research institutions; school districts; the business community; and state departments of education. CETP sought to improve undergraduate teacher preparation classes to through better collaboration among institutions and between education and science within institutions.

CETP Program Evaluation Background

Initially, NSF required each CETP project to conduct its own project-level evaluation. Several years later, building upon the recommendations from two meetings of the CETP project Principal Investigators (PIs), NSF funded a national, overarching program-level evaluation effort through a $990,000 three-year grant to PI Frances Lawrenz at the University of Minnesota.

The core evaluation goals were: (1) to achieve consensus on evaluation questions, (2) to develop, pilot, and field test evaluation instruments tied to the questions, and (3) to collect data to document CETP accomplishments nationwide.[vi] NSF required the core evaluation services to be available to all CETP projects, but it did not require the CETP projects to participate or to share data with the Core. Because involvement was optional, the Core relied on incentives to encourage individual CETPs to participate in the core evaluation process.

CETP Core Evaluation Questions

In October 1999, the core evaluation team sponsored a meeting for the CETP projects that had been funded for at least 3 years, i.e., "mature" CETPs. At that meeting, the project principal investigators reached consensus on four evaluation questions:

1. To what extent did the CETP program impact the collaboration and focus of university faculty on instructional issues?
2. To what extent did the CETP program impact the instructional techniques used by university faculty?
3. Do K-12 teachers who participated in CETP projects view their preparation programs differently from teachers who participated in other preparation programs?
4. Are the instructional practices exhibited by K-12 teachers who participated in CETP projects different from the instructional practices exhibited by teachers who participated in other preparation programs?

CETP Core Evaluation Design

Instruments. To answer the evaluation questions using project-level data, the CETP core evaluation called on local projects to develop shared data collection instruments.[vii] To start the process, the core team met with mature CETPs and gathered instruments from the local projects, as well as from Horizon Research Inc., the Third International Mathematics and Science Study (TIMMS), and the National Assessment of Educational Progress (NAEP). The CETP projects were invited to critique the instruments and to contribute to the development of the new, shared instruments. However, due to time constraints, the core evaluation accomplished the bulk of the work.

CETP Program Evaluation Facts

Funding: NSF funded $990,000 three-year grant

Activities: Core created common instruments and provided technical assistance to the individual CETPs.

Voluntary: Participation in the evaluation was voluntary.

Uniformity: Different CETPs provided different types of data so the numbers of projects providing any one type of data varied.

In January 2001, the Core convened a meeting of the CETP projects with fewer than three years of funding. Feedback from meeting participants showed their enthusiasm about the instruments and their intent to use them in their evaluations.[viii] After this meeting, the core team pilot-tested and revised the instruments. They also developed a comprehensive sampling plan and a list of responsibilities for the individual CETP projects.

The core instruments included a classroom observation protocol, a teacher artifact scoring rubric to gather information on the impact of the CETP program overall, and a series of surveys targeting either K-12 or higher education. The surveys measured the level of collaboration among higher education institutions and the amount of use of various instructional techniques before and after the CETP project. Higher education instruments were mostly web-based and included surveys for PIs/evaluators, deans/department chairs, faculty surveys (two forms), and college students (two forms). The K-12 instruments were paper-and-pencil and were tailored for principals, teachers, and students in grades 6-12 (two forms).

Data Collection. The CETP core evaluation offered two major incentives for projects to use the shared data collection instruments, in addition to general interpersonal support: (1) funding for

the effort required to collect data, and (2) cleaning and specifying all data from individual CETPs and returning it to the local projects in SPSS system files so that they could conduct their own analyses. Despite these incentives, both years of data collection yielded mixed results in that the data collected were spotty and eclectic, thus not helpful in supporting meaningful program-level conclusions.

Although NSF encouraged all 19 CETP projects funded from 1993 to 2000 to participate in the national impact evaluation, participation in the evaluation was voluntary, and in the end only 12 of the 19 CETPs actually provided usable data. Different CETPs also provided different types of data so the numbers of projects providing any one type of data varied.

Discussion of the CETP Core Evaluation

The core evaluation design prioritized collaboration and provided many opportunities and incentives for involvement. Core activities to promote collaboration included:

- Setting up a website set up immediately upon receiving funding in September, 1999;
- Hosting meetings and covering travel and participation expenses so projects could provide input on instrument development[ix];
- Providing training and materials to support the use of the shared instruments;
- Offering technical assistance and financial support for data collection[x]; and
- Creating clean, analysis-ready data sets for the projects to use in their own analysis.

In the design phase, individual CETP projects participated and expressed a willingness to help develop and ultimately to use the shared instruments for data collection.[xi] However, despite the Core's many incentives for participation, the substantial challenges of adding the core evaluation to their existing project evaluations rendered the incentives largely ineffective. Thus, much of the work of developing and refining the data collection instruments was conducted by the Core itself.[xii] In the end, several CETPs collected data using only portions of the core instruments; while others collected all the data, but from only some of their project sites.

Ultimately, given the incomplete coverage of the data, PI Frances Lawrenz questioned whether the evaluation should continue. Concerned that the CETPs were not responding to the generous incentives – paid travel and consulting time, paid time to cover data collection activities, and the offer of cleaned data delivered in a user-friendly SPSS format – Lawrenz suggested to NSF that the core project be discontinued on the basis of the poor response. Not wanting to discontinue this activity, NSF sent a letter to CETPs communicating the value of the core evaluation project. They also added participation in the Core as a suggestion for those projects applying for the additional three years of funding.

A final chapter in the core evaluation saga occurred during the final year of the evaluation's funding. An accounting error occurred resulting in PI Lawrenz being informed that funding was running low for the project. Lawrenz quickly wrapped up data collection and analysis activities and did not schedule the anticipated project meeting in which the final results were to be presented. Instead, the projects were emailed copies of the final results of the project. The

following spring the error was discovered and additional funds were made available to Lawrenz and her team. The funds were used to conduct additional and more complex data analyses and write a journal article on the final results.

Evaluation in Retrospect

The core evaluation created common instruments and provided technical assistance to the individual CETP evaluators to facilitate the evaluation of these complex projects within the short evaluation period. The core evaluation staff was also an intermediary between NSF and the CETP projects. To the surprise of the Core team, the earlier expressions of need for technical assistance and evaluation help by the projects did not result in high levels of participation by the CETPs.

Widespread participation in the early Core meetings did not translate into uniform collection of the data using the common instruments or uniform data sharing and reporting. For example, some CETPs submitted experimental data, while others submitted anecdotal data; some CETPs trained observers and conducted classroom observations in one way, some in another; some collected data only from one subset of their project participants (for example, principals, but not teachers, or teachers, but not principals). In the end, the Core sent evaluation reports to NSF and to the CETP projects, but they refrained from wider dissemination due to these flaws. Overall, the Core meetings influenced the learning of participants through discussions of instrumentation and research design. Such discussions were reportedly quite useful to the new or less experienced participating evaluators.[xiii] The core evaluation also negotiated with NSF on behalf of the projects serving as a united voice for the CETPs in approaching NSF.

NSF Program Officer's Perspective on the CETP Program Evaluation

This section of the case study examines one NSF program officer's perspectives on the agency's involvement in and use of the CETP core evaluation. Additionally, the program officer's beliefs about how use and involvement played out at the project level are presented. We acknowledge that this data is based on the perspective of only one of several program officers. It is possible that other program officers may have stressed different points or ideas.

Why was the CETP core evaluation funded? The NSF program officer interviewed for this case study explained that the core evaluation was funded to provide greater information about how the CETP program was impacting teacher preparation nationwide. A CETP program evaluation had been funded prior to funding the core evaluation conducted by the University of Minnesota, however, the program officer felt that it "didn't really tell the full story" about the program, since it was limited to the first two cohorts of projects and was conducted at an early stage in the program. Instead, NSF believed that if the projects collaborated on the development of instruments and evaluation plans, the resulting evaluation would be more "meaningful," i.e., that it would enable data to be collected across projects and would better capture the overall CETP program's impact. To enable projects to conduct longitudinal studies of the impact on faculty and students, follow-on grants were awarded to most of the

systemwide, multi-institutional projects and these projects were invited to participate in the core evaluation project.

How was NSF involved in the core evaluation? To support core evaluation activities, NSF provided the venues for planning meetings and some guidance in the selection of evaluation questions. However, the NSF program officer wanted the evaluation to be viewed as separate from NSF and as something that was important to the projects and the "field." The program officer stated that she was careful to stress that the Core, although funded by NSF, was "external to NSF and… that you were [the Core was] the one running the project, not NSF." In fact, NSF sent two Einstein program fellows, rather than NSF employees, to attend core planning meetings to prevent the projects from feeling that NSF was too involved in the evaluation.

When the first year of core funding was completed, however, and not all of the projects had provided data to the Core, the NSF program officer decided it was necessary to be more explicit about their interest in project participation in the core evaluation. In the follow-on grant announcements, consequently, NSF stated that they made it clear that participation in the core evaluation was encouraged. Still, participation was voluntary in that there were no formal consequences for not participating fully in data collection activities. However, from the NSF program officer's perspective, had they not been active in encouraging project participation, the evaluation would not have been successful. As the program officer stated, "The evaluation would have gone nowhere had we not really compelled people to participate as best as we could."

What did NSF use? The NSF program officer reported that they made some use of the core data, primarily in support of reporting requirements rather than informing decisions about the program. The primary use of the Core data was in the selection of "nuggets" – short paragraphs that highlighted something important going on the program – that were used in support of Congressional budget formulations and to meet GPRA requirements. Additionally, NSF presented the evaluation findings to the committee of visitors that reviews each NSF program every three years. The NSF program officer also believes the instruments produced by the Core are valuable resources and, to this day, refers individuals to the University of Minnesota website where the instruments and manuals are available.

What is the NSF program officer's perception of involvement and use at the project level? The NSF program officer interviewed for this study felt that the projects were involved with the core, had influence over the evaluation, and used the core data to some extent. The program officer stated that she received positive feedback from a number of the project evaluators and leaders over the course of the core evaluation. In terms of their involvement, she believed that the projects were most involved in developing the data collection instruments – by clarifying and rejecting items. As a result of their involvement, the program officer felt that the projects felt a greater sense of ownership over the core data and therefore were more likely to use the data.

The NSF program officer felt that the projects did make use of the processes and products of the Core. The program officer reported that the projects used data supplied from the Core in their own project evaluation reports to NSF. Additionally, the program officer believed that the core instruments made the projects gather higher level data about their project's impact. As the program officer said,

> I think it probably pushed them [the CETP projects] to get data that they would not have otherwise. I mean they weren't getting it in their regular, project-level evaluation … They were basically used to kind of counting people and collecting data at that level.

A second outcome of the evaluation, from the program officer's perspective, was forming a community of CETP project leaders and evaluators. Taking part in developing instruments and gathering data for the core contributed to group cohesiveness. As the NSF program officer stated, "I think this may have given a feeling that they were part of the group, contributing to a large project." And, "I don't think it was until this project [the core evaluation] came that they [local evaluators] actually became colleagues, you know partners." Additionally, the program officer perceived that the projects began to feel that the Core team members were their colleagues as well.

The NSF program officer felt that the use and influence of the evaluation was limited to some extent, however, because the core was not funded until the program had been in existence for several years. As she stated,

> I think we start too late on these projects, that's the main thing. …the involvement should be really much earlier in place. At the time we're designing a program we ought to say this is part of it; we're setting aside money that we can develop this core evaluation.

In sum, the NSF representative felt that the projects were involved, the instruments and data were used somewhat, the boundaries of evaluation were pushed forward, and a community was developed among STEM evaluators and leaders.

Project Level Involvement and Use

This section of the case study examines the relationship between project involvement in the program evaluation and the use of the evaluation process and outcomes. It presents Project Leader and Evaluation Survey and interview findings organized according to the following topics:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects

- Factors Affecting Use

For each topic, quantitative results from the online survey will be discussed first. Data from project interviews will be addressed next. Relevant examples from the open-ended online survey items may also be included.

Motivations for Involvement

Project PIs and evaluators became involved with the core evaluation in a variety of ways. Some CETP project leaders and evaluators reported that they first became involved when they responded to a general call from NSF to CETP projects. Others reported that they had worked with Lawrenz previously on other projects or that she had providing consulting services to their CETP projects. Some project leaders and evaluators mentioned that Lawrenz contacted them personally and asked them to become involved with the Core by attending one of the early planning meetings. Lawrenz was known to many project leaders and evaluators from the mature CETPs, as she had met them during the "Corridors for Collaboration" meeting hosted in Virginia in 1998.

Table 4. Motivation for Involvement in the CETP Core Evaluation

Survey Question: Which option best describes the motivation for your involvement during the:	**I was required**	**I was asked**	**I volunteered**	**N**
evaluation planning stage (chi-square = XX, df = XX, p<0.00X)	20.5%	59.0%	20.5%	39
evaluation implementation stage (chi-square = XX, df = XX, p<0.00X)	32.6%	41.3%	26.1%	46
communication of evaluation findings stage (chi-square = XX, df = XX, p<0.00X)	35.3%	35.3%	29.4%	34

Intellectual curiosity or an interest in improving teaching on their campus was mentioned as playing a role in motivating some project leaders and evaluators to become involved with core evaluation planning and data collection activities. Others, like this project director, cited multiple motivations for getting involved in planning the core evaluation.

> One, we were eager to find out about our impact, of course, and to begin to think about how to improve what we were doing because we wanted it to have a lasting effect. One of our things from the get-go was about institutionalization and making sure there was a lasting positive impact both on our students and on our faculty. So that was a big driving force. The other one was to have quantitative and qualitative data that would help us to move forward with some of the, what I would call, "spin off" issues from the project. And a third was intellectual curiosity. (CETP Interviewee 1)

Another project PI expressed similar feelings when asked what motivated her to get involved in the CETP core evaluation. Although she seemed to be responding to her motivation to get an

NSF CETP grant for her project, the research aspect of the evaluation appears to be part of her motivation. As she stated, she was motivated by her

> …interest in improving teaching on my campus. I was made aware of teaching methods that were out of date that I was using, initially, from my colleagues… It was very intriguing to me that, here we are a huge university and we're teaching in an inefficient way. So I got interested in it and once it got my attention then the funding opportunity came up within a year or two… and we succeeded in getting the funding and that propelled us into a more active phase of changing our university. (CETP Interviewee 2)

Receiving additional funding from NSF to support Core data collection activities was also cited as part of the motivation for participating in the core evaluation. As one interviewee stated, "I suppose even without the extra funding that we received, we were being asked by Core and the CETP project to contribute data, and we would have made an effort to do so. But we did feel a particular obligation because of the funding." (CETP Interviewee 5) Although all CETPs received some additional funding for providing data to the Core, only one of the interview respondents shared this motivating factor.

Survey data support the idea that most project leaders and evaluators felt that their participation in the Core was voluntary in nature. As shown in Table 4, almost 60% of survey respondents stated that they were asked to be involved in the planning phase of the core evaluation, and another approximately 20% reported that they volunteered to be involved. Only 20% reported that they were required to be involved with evaluation planning activities. It is interesting that even 20% responded that they were required, as there was no formal requirement by either NSF or the Core to be involved in any components of the evaluation. This may reflect internal pressures placed on evaluators or staff by PIs or others to take part in these activities.

A somewhat greater percentage of survey respondents reported feeling required to take part in the implementation phase and communication phase of the evaluation. In both phases approximately one third of respondents reported that they were required to be involved while the remaining two thirds reported being asked or volunteering.

Levels of Involvement

As described in the background information for the case study, project PIs and evaluators were involved primarily in two stages of the core evaluation. First, project leaders and evaluators from mature CETPs were asked to become involved in the planning and instrument development phase of the evaluation. Second, all CETPs were asked to gather data and provide it to the Core. Survey data from project leaders and evaluators provide a more detailed description of involvement in the core evaluation by project leaders and evaluators. As shown in Tables 5-7, the majority of respondents reported being involved at least a little or some in all of the aspects of evaluation planning, implementation, and communication, although between 22% and 48% of respondents reported that they were not involved in any individual activity.

As was expected, the highest relative levels of involvement were reported in collecting data (mean=2.8[xiv]) and the discussions that focused the evaluation (mean=2.8), indicating slightly below the level of "some involvement." Over three quarters of survey respondents reported being involved at least a little if not extensively in the discussions that focused the core evaluation. High percentages of project leaders and evaluators also reported being involved in developing data collection instruments, developing data collection processes, and collecting data. Between 70% and 75% of respondents reported being involved at least a little in these activities.

Table 5. Involvement in evaluation planning activities

Survey Question: I was involved in…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
the discussions that focused the evaluation.	22.6%	13.2%	30.2%	34.0%	53	2.8	1.2
developing the evaluation plan.	35.2%	16.7%	24.1%	24.1%	54	2.4	1.2
identifying evaluation planning team members.	42.6%	18.5%	16.7%	22.2%	54	2.2	1.2

Somewhat fewer survey respondents reported being involved in some way with reviewing, analyzing, or interpreting the data. The lowest involvement levels were found for developing future plans (mean=2.1), analyzing data (mean=2.2), identifying evaluation planning team members (mean=2.2), and writing evaluation reports (mean=2.2). For these items, approximately 40-45% of respondents reported that they were not involved at all.

Table 6. Involvement in evaluation implementation activities

Survey Question: I was involved in…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
collecting data.	24.1%	9.3%	25.9%	40.7%	54	2.8	1.2
developing data collection processes.	28.3%	17.0%	26.4%	28.3%	53	2.6	1.2
developing data collection instruments.	30.2%	17.0%	26.4%	26.4%	53	2.5	1.2
reviewing collected data for accuracy and/or completeness.	43.4%	7.5%	18.9%	30.2%	53	2.4	1.3
interpreting collected data.	39.6%	17.0%	15.1%	28.3%	53	2.3	1.3
analyzing data.	43.4%	18.9%	17.0%	20.8%	53	2.2	1.2

Table 7. Involvement in evaluation communication activities

Survey Question: I was involved in…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
reviewing evaluation reports for accuracy and/or completeness.	37.0%	14.8%	18.5%	29.6%	54	2.4	1.3
presenting evaluation findings.	38.9%	14.8%	24.1%	22.2%	54	2.3	1.2
writing evaluation reports.	48.1%	11.1%	13.0%	27.8%	54	2.2	1.3
developing future project plans based on evaluation results.	44.4%	18.5%	18.5%	18.5%	54	2.1	1.2

Involvement in Evaluation Planning and Instrumentation. Some of the project leaders and evaluators who were interviewed discussed their early involvement in the development of the Core by contributing to the evaluation and instruments for conducting the evaluation. As described by one interviewee,

> I was involved in the early stages with [my colleagues] in responding to the whole notion of developing a phase two. And we actually participated, [my colleague] and I did, with Frances and a number of other key people to begin to develop the instruments. We really carved out the vision for what the core evaluation would be and what it would look like and what kind data we might collect. (CETP Interviewee 1)

Another project PI also discussed how her project was involved in the planning and design phase of the core evaluation. She stated that,

> …for the design part, Frances did consult with me and my team about the instrument that she used. It was quite different from ours, but she did consult with us on the use of [our project's] evaluation instrument, which we used as a classroom evaluation tool. It was a little checklist. We had some conversation with her about that. So I would say that our team did participate with her, collaborate with her on the design of the evaluation. It was mostly my colleagues who had the expertise. (CETP Interviewee 2)

Several individuals cited attending planning and training meetings in Minnesota when asked about how they were involved in the core evaluation. One respondent felt that she was not involved in the core evaluation until she attended two of the meetings in Minnesota. However, her experience at the meetings was mixed. Although she reported enjoying the networking opportunities provided by the gathering, she found the task of developing data collection instruments to be difficult. As she stated,

> We also found the meetings to be quite daunting as we talked through the instruments that Core had designed and we were expected to use… I remember thinking at the first

> meeting that the instruments were really poorly designed, partly I think that was because of the uniqueness in [our CETP], but it felt as though, for example, with the student evaluation instrument, it felt as though it had been written by people who hadn't spent very much time in K-12 schools. And so we worked hard on that, my project manager and I, and everybody else did, too, and I think we made significant improvements on the instrument and fine tuned those again at the next meeting. (CETP Interviewee 5)

Other projects' leaders and evaluators also reported becoming involved with designing the instruments as part of attending the meetings, although these individuals did not express the same level of frustration with collaborating on their development. As one interviewee shared,

> We would meet annually, [Lawrenz] would have people come back to Minnesota. And we would discuss, but it was really early on, we would discuss how things were going. But I don't recall much feedback in terms of the results. Actually, I think the first couple of years we were helping to make the surveys up… I remember helping to make up the survey, for sure. I mean adding questions and things like that. And then the next year looking at maybe preliminary data. (CETP Interviewee 3)

Data Collection. Providing data to the Core was an additional way that many of the projects reported that they were involved in the core evaluation. The interviewees mentioned that they complied with Lawrenz's requests for data and sent in data from their projects to the Core. The data collection process was one that one project PI reported as being a large burden for her project, however. As she shared,

> We found that the expectation of sending out the student evaluation forms to teachers in classrooms, both those that had and had not been impacted by the original CETP project, and trying to get administrative responses and trying to get faculty responses was really a daunting task. And it used an immense amount of time… We also found that the procedure was not at all convenient. For example, rather than doing the duplication of the actual hard copies of the instruments ourselves, we were to identify the names of the individuals who would participate and their address and email and stuff…. Rather than make the duplications ourselves, we were sent boxes of multicolored forms to be distributed then by us. And it was a real complex procedure that we felt we could have streamlined. (CETP Interviewee 5)

When asked how involved this individual felt she was in the core evaluation, she responded, "I would say that we, my colleague and I, were involved extensively because of the amount of time and energy we put into collecting data." (CETP Interviewee 5) It is important to note that none of the other interviewees shared similar feelings that the data collection process was particularly burdensome.

None of the individuals interviewed for this case study mentioned involvement in later phases of the evaluation such as analyzing data, writing reports, or disseminating evaluation findings when asked about ways in which they were involved with the Core.

Perceived Effect of Involvement

The vast majority of survey respondents indicated that they felt their input was valued. When asked on the survey if they felt that the input they provided in the core evaluation was taken into consideration during the three stages of the evaluation, only one or two survey respondents indicated "no," while the remaining responded that they felt their input was considered. Similarly, only one or two individuals indicated that they felt that their input was not used, while the vast majority felt core evaluation team members used their input. Table 8 provides further details.

Table 8. Consideration and use of project leader and evaluator input

Survey Question: The input I provided during the…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N***
evaluation planning stage was taken into consideration.	1.9%	14.8%	33.3%	18.5%	37
evaluation implementation stage was taken into consideration.	1.9%	16.7%	35.2%	22.2%	41
communication of evaluation findings stage was taken into consideration.	3.7%	14.8%	29.6%	14.8%	34
evaluation planning stage was used.	1.9%	24.1%	25.9%	13.0%	35
evaluation implementation stage was used.	1.9%	16.7%	35.2%	16.7%	38
communication of evaluation findings stage was used.	3.8%	18.9%	24.5%	13.2%	32

*Total N is small because some individuals responded that they did not provide input or they did not know if their input was considered or used.

Although some project leaders and evaluators stated during their interviews that they were uncertain if the feedback they provided to the Core was considered and used, several indicated that they felt it was valued. As one project director responded, "Oh, I think it was used. It was considered certainly, and I think it was helpful." (CETP Interviewee 2) Another commented that

> I think that when we were there [at the planning meetings], we felt that we were all heard, all listened to. It seemed like a much larger task than we could handle (laughs). But there were clearly things that had to be cut out or changed depending on the institution. (CETP Interviewee 1)

Use by Project Leaders and Evaluators

*Survey findings***.** The online survey asked project leaders and evaluators to reflect on the extent to which the CETP core evaluation influenced their knowledge and understandings, skills, and beliefs about evaluation. Additionally, project leaders were asked about their use of the core evaluation process and findings to make decisions about their project or in other projects. As shown in Table 9, self-reported data from the survey suggest that the CETP core evaluation had the greatest influence on project leaders' and evaluators' knowledge and understandings, followed closely by influences on their skills, beliefs, and use in other contexts. To a lesser extent, Project PIs used the core evaluation in a variety of ways including planning future projects or seeking funding for the project (which is discussed below in the Use by Projects section). It must be reiterated that a non-response bias was found for these evaluation "use" questions. The data suggest that CETP survey respondents reported higher levels of use and influence of the core evaluation than individuals who did not respond to the survey. Therefore, it is likely that the levels of use reported below are somewhat higher than the true use levels in the CETP project population.

Table 9. Influence of core evaluation on project leaders and evaluators

	N	Mean	SD
Influence on knowledge/understanding (q25-30)	54	2.6	0.9
Influence on beliefs (q38-43)	54	2.5	1.0
Use of knowledge, products, processes in other contexts (q45-49)*	31	2.5	1.0
Influence on skills (q31-36)	53	2.4	0.9
Ways of using findings (q50-58)**	27	2.0	0.8

*items asked only of those reporting having conducted a subsequent evaluation
**items asked only of project PIs

Influence on Knowledge and Understanding. Over 70% of all project leaders and evaluators who responded to the survey reported that the core evaluation increased their knowledge or understanding of evaluation stages, STEM education, STEM education evaluation, and their project at least to a little extent. Approximately a quarter of respondents indicated the evaluation influenced their knowledge and understandings extensively. As shown in Table 10, survey respondents indicated that the evaluation had the greatest positive influence on their knowledge and understanding of their project (mean=2.8), STEM education evaluation (mean=2.7), and how to implement an evaluation (mean=2.7).

Table 10. Influence of the CETP core evaluation on project leaders' and evaluators' knowledge and understanding

Survey Question: The evaluation increased my knowledge/understanding of…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
my project.	16.7%	14.8%	44.4%	24.1%	54	2.8	1.0
how to implement an evaluation.	18.5%	11.1%	50.0%	20.4%	54	2.7	1.0
STEM education evaluation.	14.8%	24.1%	38.9%	22.2%	54	2.7	1.0
how to plan an evaluation.	25.9%	11.1%	40.7%	22.2%	54	2.6	1.1
Science, Technology, Engineering, and Mathematics (STEM) education.	27.8%	13.0%	37.0%	22.2%	54	2.5	1.1
how to communicate evaluation findings.	27.8%	20.4%	33.3%	18.5%	54	2.4	1.1

Influence on Skills. Like the responses to questions about their knowledge and understandings, almost 70% or more of respondents felt the core evaluation had at least a little influence on their evaluation and STEM education skills, although smaller percentages reported extensive influence in these areas (11-19%). Again, as shown in Table 11, improving the leaders' and evaluators' skills for working on their project (mean=2.6), skills as a STEM education evaluator (mean=2.5), and skills in implementing an evaluation (mean=2.6) were rated highest.

Table 11. Influence of the CETP core evaluation on project leaders' and evaluators' skills

Survey Question: The evaluation improved my skills…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
in implementing an evaluation.	18.9%	18.9%	45.3%	17.0%	53	2.6	1.0
for working on my project.	18.9%	22.6%	43.4%	15.1%	53	2.6	1.0
as a STEM education evaluator.	18.9%	28.3%	34.0%	18.9%	53	2.5	1.0
in planning an evaluation.	32.1%	13.2%	35.8%	18.9%	53	2.4	1.1
in communicating evaluation findings.	32.7%	21.2%	32.7%	13.5%	52	2.3	1.1
as a STEM educator.	32.1%	32.1%	24.5%	11.3%	53	2.2	1.0

Influence on Beliefs. The influence of the core evaluation on project leaders' and evaluators' beliefs was somewhat lower than the first two areas, with approximately one third of all respondents indicating that there was no influence. As indicated in Table 12, the survey responses suggest that the core evaluation had the greatest influence on leaders' and evaluators' belief in the importance of planning an evaluation (mean=2.7) and implementing an evaluation (mean=2.6).

Table 12. Influence of the CETP core evaluation on project leaders' and evaluators' beliefs

Survey Question: The evaluation increased my belief in the importance of…	No	Yes, a little	Yes, some	Yes, extensively	N	Mean	SD
planning an evaluation.	25.9%	11.1%	31.5%	31.5%	54	2.7	1.2
implementing an evaluation.	28.3%	11.3%	32.1%	28.3%	53	2.6	1.2
communicating evaluation findings.	33.3%	9.3%	29.6%	27.8%	54	2.5	1.2
STEM education evaluation.	29.6%	20.4%	24.1%	25.9%	54	2.5	1.2
my project.	25.9%	24.1%	37.0%	13.0%	54	2.4	1.0
STEM education.	35.8%	26.4%	22.6%	15.1%	53	2.2	1.1

Use in Another Context. Individuals who reported being involved in an evaluation since their involvement in the CETP project were asked whether they had used the knowledge, products, or processes that arose from the core evaluation in another evaluation context. As shown in Table 13, over four fifths of respondents indicated that they had used what they learned from planning and implementing the evaluation in another evaluation context. Interestingly, almost 60% of respondents indicated that they had used the program evaluation as a model for another evaluation, and two thirds of respondents stated they had used data collection instruments developed for the CETP core evaluation in another evaluation. An example of this type of use is illustrated in this open-ended response to the survey. The respondent stated that "two doctoral students I have advised recently have selected the CETP adaptation of the Classroom Observation Protocol as one tool for data collection in their dissertation research."

Table 13. Use by project PIs of knowledge, products, processes in other contexts

Survey Question:	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N**	**Mean**	**SD**
I used what I learned from planning the CETP program evaluation in another evaluation.	16.1%	22.6%	29.0%	32.3%	31	2.8	1.1
I used what I learned from implementing the CETP program evaluation in another evaluation.	16.1%	22.6%	35.5%	25.8%	31	2.7	1.0
I used data collection instruments from the CETP program evaluation in another evaluation.	32.3%	12.9%	29.0%	25.8%	31	2.5	1.2
I used what I learned from communicating evaluation findings from the CETP program evaluation in another evaluation.	35.5%	19.4%	25.8%	19.4%	31	2.3	1.2
I used the evaluation plan from the CETP program evaluation as a model in another evaluation.	41.9%	16.1%	22.6%	19.4%	31	2.2	1.2

Interview Findings. During the interviews, project leaders and evaluators provided more detailed information about the ways in which the evaluation increased their knowledge and skills and also identified one additional area in which the evaluation had influence –in building connections between project leaders and evaluators in different CETP projects around the country.

Lessons Learned about Planning Evaluations Early. First, two interviewees stated that they learned about the importance of planning for evaluation early in a program's development after seeing the effects of implementing the core evaluation after many of the projects had already been funded for several years. As one project person said after describing how it was difficult to have an evaluation begin late in her project's funding cycle, "I think the bottom line is to think about the evaluation up front" (CETP Interviewee 1). Another person explained that,

> For evaluators, the idea is that planning is a big part. So, from that standpoint, what I learned was the importance of planning ahead of time. And I always felt, looking back on it, that data gathering immediately was important. So those questions, really, were

> good questions that they had in the Core, but I wished they had been there early on. (CETP Interviewee 3)

Two individuals responded similarly in the open-ended comments on the survey: "For our case, the fact that we were not included in the discussions of the evaluation early in our grant hindered our ability to influence the direction of the evaluation design and our ability to gather any useful interpretations from it." Another wrote, "The central evaluation was not announced or planned when the project was funded and initiated. Superimposing it later compromised the original evaluation because extra resources were not given to accomplish both efforts."

Broadening of Project Leaders' and Evaluators' Perspectives. CETP project leaders and evaluators reported that they valued learning about issues facing their colleagues at other CETP projects around the country during the core evaluation meetings they attended. Several reported feeling that they acquired a broader perspective about issues facing the STEM education field as a result of their attendance at the meetings. As one project PI shared,

> Anytime that Frances came to those meetings, it was very useful because she gave us that overview, that perspective of all the CETPs and what they were doing. Otherwise we were just isolated little people in our own institutions, with our own institutional problems. And it really broadened our viewpoints and perspectives. (CETP Interviewee 2)

Another project director explained that she has been able to translate what she learned from her attendance at planning meetings into a different context in a subsequent position. As she explained, at the meetings she learned

> the very nitty-gritty details of each collaborative and what they needed and what they could and could not do… For example, one of the things that we had difficulty with in [our collaborative] was actually to get into classrooms and getting classroom data because of all the legal issues and what not. And that didn't seem to be such an issue with some of the rural areas. I am now, for example, working [elsewhere in the country] where we often have rural, urban, and suburban school districts that we work with. So I think for me in particular it was a really good experience to be part of the collaborative planning of those evaluation instruments. Like right now I'm working on a grant proposal with them and it's had a lasting effect. (CETP Interviewee 1)

This individual also shared that she learned about the complexity of conducting a program evaluation on such a large scale. She explained that she grew intellectually by being part of the whole instrument development process,

> I think that one of the lasting impacts on me was how different the CETPs were and how data collection could proceed in somewhat the same way in all of them, and yet had to be tailored to each specific region or state or urban area. It was quite a design test… and when we do collaborative work, it's messy. Some times we have to make tough

> choices and not do things that we want to do. That was a lasting effect on me: how complex the whole thing could be. (CETP Interviewee 1)

Learning New Evaluation Skills. Two of the project evaluators interviewed for the case study stated that they felt they learned new evaluation skills by participating in the evaluation, despite one being new to the field and one being at the end of his career. As the first stated, "I think the more questions I see and the more evaluation instruments I see - certainly it has had some general impact… This was my first evaluation so it just gave me more experience."(CETP Interviewee 4) The second shared that he learned new methods, however, wasn't able to apply them because he was at the end of his career: "I know at the time I was very interested in survey construction at the time and more methodology. And I thought at the time that different methods of analysis, like Rausch analysis, were appropriate. But I was at the end of my career… so we felt it was time to end it, time to move on." (CETP Interviewee 3)

Building a Network of CETP Project Managers and Evaluators. In addition to building their knowledge and skills, several of the interviewees reported that they valued the relationships they built with other CETP project leaders and evaluators as a result of attending the core evaluation planning meetings. One project PI stated that she felt that "the process of Frances doing the evaluation was useful to me because it gave me a perspective overall of the projects. And so I could see what other people did on their campuses and got to communicate with them." (CETP Interviewee 2) Similarly, another PI stated that she found the meeting she attended to be "useful in that it helped us a great deal in establishing relationships with other CETPs' project directors and learning the types of activities they were involved with, so they were helpful in terms of networking and collaboration." (CETP Interviewee 5)

Another felt that she could contact other CETP projects although she has not yet had the need: "I could call the Colorado people or the Montana people or something, you know. I could say "hey, remember me?" That's always a possibility; I just haven't had the opportunity or the need to do that." (CETP Interviewee 1)

One open-ended response on the survey also indicated that the meetings were helpful in building connections between project PIs and evaluators: "The opportunities to meet and share with other evaluators, especially when we met at the University of Minnesota, was outstanding. I benefited a great deal from the professional discussions and Frances' leadership."

Not All Reported Positive Influences. Not all project leaders and evaluators felt their participation in the core evaluation was a positive experience. One interviewee in particular felt that the process was highly frustrating and that she didn't gain anything from her involvement with the Core. She stated that the core evaluation caused her "a lot of frustration" due to a lack of understanding by NSF and the core evaluation team about the particularities of her project and how the evaluation design did not fit her context. When asked if she learned anything new about evaluation as a result of the experience, she responded, "We already had evaluation experts already involved in the program and I had extensive experience, so I don't think we gained anything." (CETP Interviewee 5)

On the survey, project leaders and evaluators were asked if the evaluation had a negative impact on their beliefs about evaluation. Of the 52 respondents to the question, only seven or 13% indicated that the evaluation did have a negative impact on their beliefs about evaluation. When asked to elaborate, survey respondents shared two types of concerns. First, a few were concerned that the evaluation results were not taken into consideration when making decisions. For example, one respondent stated that "the evaluation contributed to my frustration regarding working with a client that did not appear to be interested in using evaluation results." Another wrote that there was a negative influence "only to the degree that evaluation data and analysis was not always considered when programmatic decisions were made. Political, practical, logistical and economic considerations often took precedence over the evaluation results when planning and implementing project activities."

Second, a few survey respondents, like the interviewee quoted above, indicated that they had concerns about the design of the evaluation and the instruments used by the Core. For example, one respondent shared that the "CETP projects were quite diverse, and each had already designed its own evaluation. The overall, NSF evaluation compared apples and oranges (which negates validity), and in the process stifled the local evaluation efforts." Another wrote that "I was especially negative about the process by which the evaluation was constructed and the instrument produced by this process."

Use by Projects

In addition to asking questions about the individual impact of the CETP core evaluation (addressed in Use by Project Leaders and Evaluators above), interview respondents were asked to comment about the impact of the core evaluation on their projects. Few of the individuals interviewed for this case study reported that the core evaluation had much of an impact on their project. Most reported, instead, that the project-level impact was minimal or none. For example, one responded to a question about the level of impact on the project as such: "I would say almost none… it's hard to say there was no influence, but from a timeline standpoint looking at this from outside, you would say 'How could there be?'" (CETP Interviewee 4) As this individual's project was in its final year of funding when the Core was established, it is understandable that the impact on the project was minimal. Other reasons for the lack of project-level impact are discussed below.

One project director was unable to speak to the actual long-term impact of the evaluation as she left the project many years ago. However, she believed that if the courses they had developed as part of her CETP project were still in place that would be a strong indication of the positive impact of the project. Similarly, if the data that they collected as part of the core evaluation had helped to develop better programs and to obtain future funding, she felt this would be an important lasting effect of the core evaluation. As she said,

> I would hope that the data that our group was able to get [from the core evaluation] and use would then be used subsequently to help the project develop new programs

> and get new grants. And look at the sustained impact of the project over time, because I guess that's one thing that we don't really have a sense of. You know, how long does this last? Before we have to reinvent something, or start it up again. So there is this kind of idea of residual impact - is it five years, four years, before we have to gear it up again? (CETP Interviewee 1)

Interestingly, this interviewee characterized her level of involvement in the core evaluation as only "trivial" despite being involved in framing a vision for the evaluation. When asked about the impact on her project, she stated, "Well, I guess because I was involved at such a trivial level, I really saw it as helping us to phase out phase one and gear up and give us direction for phase two. So it really grounded us in what we had to produce and the things we had to think about, in terms of turning things over [to a new group of leaders and evaluators]." (CETP Interviewee 1)

Survey data support the idea that the impact of the core evaluation on the projects was minimal. As shown in Table 14, mean responses from project PIs regarding the use of the core evaluation findings in their projects were primarily around the "some use" level. Few reported using the Core findings in the sorts of instrumental ways one would anticipate if this were not a multi-site context.

Table 14. Ways project PIs used CETP core evaluation findings

Survey Question: I used the CETP program evaluation findings to…	**No**	**Yes, a little**	**Yes, some**	**Yes, extensively**	**N***	**Mean**	**SD**
increase the attention given to my CETP project.	37.0%	11.1%	40.7%	11.1%	27	2.3	1.1
develop future plans for my CETP project.	37.0%	22.2%	33.3%	7.4%	27	2.1	1.0
enhance my organization's commitment to my CETP project.	40.7%	11.1%	44.4%	3.7%	27	2.1	1.0
make decisions about the future existence of my CETP project (e.g., cancel or continue).	44.4%	11.1%	44.4%	0.0%	27	2.0	1.0
meet contractual and/or legal requirements of my CETP project.	46.2%	19.2%	23.1%	11.5%	26	2.0	1.1
make changes to my project(s).	42.3%	23.1%	30.8%	3.8%	26	2.0	1.0
promote additional evaluation of my project(s).	63.0%	7.4%	22.2%	7.4%	27	1.7	1.1
help plan a new project(s).	57.7%	19.2%	19.2%	3.8%	26	1.7	0.9
seek funding for my project(s).	63.0%	14.8%	18.5%	3.7%	27	1.6	0.9

*Only project PIs were asked this question, hence the lower N.

Factors Affecting Use

Project leaders and evaluators suggested a number of factors that both enabled and limited the impact of the core evaluation on the projects.

Enabling Factors

Collaboration. In the interviews, project leaders and evaluators were asked to respond to the question, "What do you think were the reasons the CETP core evaluation had the impact that it did?" Project leaders and evaluators primarily identified issues arising from the involvement of the CETP projects in the core evaluation design process as those that enabled the Core to have an impact on them and their projects. Involvement in initial planning meetings, as well as follow-up instrument training meetings, led project leaders and evaluators to feel committed to the evaluation process and confident in the data collection instruments. Additionally, meeting attendance was felt to facilitate a broadening of their perspectives on issues facing the field. As one project director stated when asked the reasons for the core evaluation's impact,

> …the way that [the core evaluation] was initially handled, how all the different projects were brought together. That there was a broad base of voices and perspectives in the initial planning group. I think those factors were both important to the success of the evaluation process as well as keeping the key players involved and committed to the process, and making sure that all kinds of projects were readily represented in the instruments that were created. (CETP Interviewee 1)

Another project PI stated that the core evaluation planning meetings,

> …tied everything together and gave us a broader perspective of the national problem, of national issues. And so it was a general problem. It enlightened us. It enlightened me. …it influenced me and my perspective and I could see that it did in my colleagues and people with new projects were influenced. So it was good. I wish she [Lawrenz] would have come along sooner in my project. (CETP Interviewee 2)

In contrast to positive remarks by the previous interviewees, the one project PI who was interviewed who had not been involved in the development of the core evaluation data collection instruments had a very different perception of their quality and usefulness. As she shared,

> In the development of the instruments… particularly in the surveys for the students, it felt to me that the developers hadn't had any experience in K-12 schools to be writing surveys like that. I can't remember specifically why, but that was my impression.

And,

> …it's not clear to me how the activities of the Core group got started. If they started with communication with PIs and classroom teachers and probably co-PIs, then I think that was an excellent way to proceed. Again, I was very concerned about the instruments when they were first developed, and I just couldn't imagine where they could have come from. They seemed to come from out of the blue from people who weren't sure of the context in which they were working. (CETP Interviewee 5)

Without engaging first-hand in the messy experience of trying to develop instruments to collect common data across disparate projects, this individual was highly critical of the surveys. This stands in contrast to those involved in their development who may have seen some flaws, but understood the complexities involved in their creation. As a result, this project submitted data from their own classroom observation protocol, rather than the Core's version.

> It was intended that we would use the Core classroom observation protocol. So we gathered together some retired classroom teachers, and we trained them in on observing, but when we showed them the Core classroom observation protocol and our [project's classroom observation protocol] they much preferred ours. So we got permission to use that and submit it. I never have figured out why we got permission to use ours, because it wasn't exactly the same type of data, and I don't see how it could have fit into the data analysis. (CETP Interviewee 5)

Open-Ended Survey Responses. A total of 32 of the 54 survey respondents provided open-ended responses to the question: "What major factors, if any, enabled the evaluation experience to influence you?" Almost half of the open-ended responses (14 of 32) cited factors relating to the project evaluators' and leaders' involvement in the planning and instrument training meetings as enabling the core evaluation to have an impact on them. Specifically, ten survey respondents indicated that they valued meeting with other CETP PIs and evaluators and collaborating in the planning of the evaluation. Another four responses specifically mentioned involvement in developing the core evaluation instruments as an important factor affecting the evaluation's influence. For example, one respondent wrote that "the face-to-face meetings with other evaluation personnel from the various CETP projects were extremely powerful. This provided context for the evaluation of my project within the whole CETP scope." Similarly, another wrote, "The face to face discussions with other CETP PIs were critical. They helped to identify the problems with the evaluation process." A project PI reported that, "My participation in assisting in the development of the observation protocols was extremely valuable both in the procedures and the outcomes."

The open-ended responses highlighted additional factors that enabled the evaluation to be influential. Seven responses discussed qualities of the core evaluation team as important to the evaluation's influence. For example, one respondent wrote about "the expertise of the PI and the organizational skills she demonstrated in presenting an effective system for evaluation [and] the PI's ability to bring people together to work on an evaluation framework. The construction of a good set of questionnaires helped not only the immediate project but helped in the development of other Project Evaluations."

As in the comment above, an additional three respondents stated that the overall quality of the instruments was also an important enabling factor. For example, "I found the instruments developed by the CETP program to be quite good and have used them as models for instruments in other evaluations."

A few respondents also mentioned that the communication structures between evaluators and PIs at the local project level was important for increasing the influence of the evaluation. As one PI stated,

> Our project's Lead Evaluator, who represented us on the CETP Evaluation planning team, regularly sat down with me and shared what was occurring on the team, and asked for my input on matters the team was considering. I also received mailings of draft evaluation instruments and processes from the team, but these sit down meetings got my attention and were a useful means for involving me in the planning and implementation process.

Similarly, another project PI wrote, "Working closely with our CETP project evaluator and implementing the Core classroom observation tool gave me a much better understanding of the methodology and how the results could be focused and triangulated to obtain very useful data/conclusions. This was fun!"

Other factors mentioned in the survey open-ended responses included the importance of the Core findings, the influence of project staff members, and support at NSF as being important factors in increasing the evaluation's impact.

Challenging Factors

Core Evaluation Timing Issues. One clear theme that emerged from the interviews and survey responses was concern about NSF's timing in implementing a core evaluation. As explained by this evaluator from a mature project, the Core was implemented late in their development with few guidelines from NSF.

> I think the attitude of many of the people who were involved early in the CETP project was that when the Core came, it was after the fact. It wasn't after the fact for other CETPs, but it was for ours. And there was a lot of negative feeling by evaluators, working and finding very little guidelines. NSF provided very little guidelines and now, all of the sudden, the projects are over, and they're saying, "Let's get these data," when in fact that should have happened early on. From my standpoint, it was more about bad planning by NSF. After the fact. And it was an attempt to try to make something out of these projects. (CETP Interviewee 3)

As suggested in the quotation above, projects were affected by the timing of the core evaluation differently depending on where individual projects were in their funding cycles. Two of the three mature CETPs interviewed for this case stated that the core evaluation was not able to have an impact on their projects because their projects were already past the formative stages and they were not able to use the data. For example, one evaluator explained that

> …the problem was that our project was pretty much over by the time the Core had started, because our project really ran until 1999 [the Core was funded in 99]. The last

> three years, until 2002, was really for the evaluators to more or less look at data, and they [NSF] still continued to fund various aspects that we thought had been successful. But in reality we couldn't make use of the Core because our project pretty much was over. (CETP Interviewee 3)

One interview respondent from a "developing" CETP project also cited timing issues as preventing her from making use of the CETP instruments. The project was in its second year of funding when the Core began and consequently had already developed its own instruments to collect project evaluation data. As she said, "I feel badly that we didn't get more advantage from the Core group, that there wasn't more involvement when we were working on our evaluation design, but I suspect that was mostly because of the timeline. But it would have been good if the Core could have helped with design." (CETP Interviewee 5)

For some projects, however, the timing of the core fit into their project's lifecycle and was consequently helpful in the project's development. As this representative from a mature CETP discussed, planning for the core evaluation helped her project transition from the first five-year grant (phase one of the project) into the second phase of the project supported by a three-year follow-on grant provided by NSF. The core evaluation helped the project

> …to phase out phase one and gear up and give us direction for phase two. So it really grounded us in what we had to produce and the things we had to think about, in terms of turning things over. Because it really was a whole different group of people in our case who were involved [in the phase two project evaluation]… So from an administrative point of view, it really was winding down and preparing for just data collection and trying to figure it out. (CETP Interviewee 1)

Open-Ended Survey Responses. Similar sentiments were expressed in the open-ended comments from the surveys. A little over a third of survey respondents (20 of 54) provided responses to the question, "What major factors, if any, limited the influence the evaluation experience had on you?" Of the twenty responses, almost one half or nine responses addressed issues that were related to the timing of the evaluation, their project, or their personal career as limiting the influence of the evaluation. As an example of how project timing affected evaluation influence, one respondent wrote, "Ours was an older project when the CETP overall evaluation began, and [I] wish we'd had these instruments and plan in place when we began." Similarly, another respondent shared that "evaluation development occurred near/after conclusion of major elements of our project so it had little influence. Would have potentially had more if more timely-so perhaps later projects felt more benefit."

Lack of Clear Evaluation Expectations by NSF. A second issue affecting the impact of the core evaluation was that NSF reportedly did not provide clear evaluation expectations for project PIs. As this person reported in an interview,

> I think they wanted a scientist to be the Principal Investigator. But they didn't warn me, or educate me, that there was something called an evaluation that was needed. And

> that there's a whole field out there. We're very narrow when we're faculty; we all study one thing. So all I knew was Astronomy. So I wasn't aware of the immensity of the field and what was involved. So I was a couple years into the project when I realized that you need a really deep evaluation. That it's really important. So my unawareness, my ignorance of that, was a problem. And I hope, perhaps, in subsequent years that scientists caught on to that. But if they're going to require that scientists be in charge of a project then I think they could have made me more aware of that. That would have been useful. So if Frances, for example, had started her core evaluation earlier, I think I would have become aware of it sooner. That's all. And it just so happened that I was one of the first. (CETP Interviewee 2)

Several other interviewees suggested the need for clearer communication between NSF and project leaders and evaluators. These two evaluators discussed that NSF expectations were unclear which caused confusion by the projects.

> I think well before the core I think that the NSF, people were concerned, that they didn't know exactly what NSF wanted. Again, I think the biggest thing was more from [the] NSF side by getting involved with evaluators and saying this is kind of what we expect. That not every project would have to be the same, but certain data would have to be similar. (CETP Interviewee 3)

> I think there were mixed messages and everybody, people weren't sure what NSF wanted, and by putting a core evaluation together early that would have been good. (CETP Interviewee 4)

Another project PI mentioned that NSF did not make clear the way that the projects were compensated for providing data to the Core, which consequently caused issues between at least one project and a Core team member. As she described,

> I was in communication pretty consistently with [our NSF program officer]... but when I was communicating with her about writing the proposal and talking about how the follow-on money might be distributed, she informed me that I could also ask for [funding] toward my project that would be sent from Frances Lawrenz as a Core grant. And so I wrote that into our proposal that we would have [funding] for three years, and the grant was funded that way. But when I explained that at some later date to Frances about how that money was going to be sent to us and what we should do to get it processed, Frances had never heard of such a thing, and it took us quite some time to work that out. In fact, at one point there was a heated discussion, between [a Core team member] and our financial clerk who didn't necessarily completely understand how NSF works, but she was trying to figure out how she was going to get the money. Eventually we did work that out. (CETP Interviewee 5)

Two responses to the survey's open-ended question addressed NSF issues, one in response to the question about limiting factors and a second as part of the open-ended concluding

question. The following response provides another perspective into internal issues affecting NSF. As this person wrote,

> The biggest limitation was the interaction between a PI who had little support from the NSF and a project officer from the NSF who drove everybody crazy by not knowing what she wanted and continually threatening to cut off funding. This led to the PI and co-PIs of the project continually relying on the evaluation team to influence the project.

It is an interesting suggestion that while perhaps the core evaluation itself had limited influence, at least from this respondent's perspective, the core evaluation team appeared to have some influence with NSF on behalf of the projects.

The second open-ended response illustrates how political decisions could cause frustration on the local level. As the following survey respondent wrote,

> It was frustrating to be collecting follow-up and impact data on the CETP project when it has already been discontinued at NSF. Political decisions were made to reallocate the CETP funds to other NSF programs before there was enough systematic evaluation results to judge the value of the project.

Relevance of Core Data to Projects. In addition to timing and NSF issues, some CETP projects did not feel that the core evaluation design fit their project, and consequently the data collected and analyzed by the Core were not helpful in making evaluative decisions about their project. As one project PI explained in her interview, because her Collaborative reached almost every pre-service teacher, the task of identifying an appropriate comparison group who were not part of their CETP project was impossible. Consequently, she explained that

> We were constantly told, this was from our program officer at NSF, that we needed to identify students who were CETPed and we needed to identify K-12 students whose teachers were CETPed. But it was really impossible. We certainly made a stab at it, but eventually, after five years, we had made such an impact on the state that it went far beyond the statistics in our activities. (CETP Interviewee 5)

Similarly, one open-ended response to the survey documented that the evaluation was reportedly too broad to be relevant to his or her project. The individual felt that the core evaluation

> …seemed marginally relevant, overly generalized… Of greater benefit were external project evaluators and visiting committee commentary and consultations. After the fact, I did not find process nor the evaluation compelling. Evaluation alignment with our project were marginally parallel.

Another PI stated on the survey that the core evaluation "marginalized our own local evaluation efforts, which would have been much more relevant to our own project."

Dissemination of Core Evaluation Findings. Several of the project leaders and evaluators in the interviews and surveys mentioned that they were not aware of any dissemination of a final report on findings from the CETP core evaluation or that the way the data were presented was not useful. For example, as this individual stated in her interview,

> We were sent, eventually, some massive reports and so I can't claim that nothing was done with the data. It obviously was analyzed. But I honestly I didn't read those reports. I skimmed them. I was curious about them. But I didn't find any executive summary, for example, that was of sufficient interest to me to find that there were any interesting conclusions drawn. So I don't know what happened to all that data. (CETP Interviewee 5)

The same interviewee mentioned that she had a third meeting for project leaders and evaluators in which the final results from the evaluation would be shared.

> I had assumed that because we had a three-year follow-on grant and the meeting was held once per year for the first two years of the grant, I assumed that we would have had a similar meeting during the third year, and by that time there would be a compilation and analysis of all of this data that everyone was collecting from all across the county, that we would all be able to draw conclusions from it, but the third year meeting never existed. (CETP Interviewee 5)

The same PI mentioned that she had "for many years been looking for published articles, but I haven't seen anything that's come out of the core. I'm not aware of anything… And I'm not aware of presentations at major science and math meetings." (CETP Interviewee 5)

Similarly, two of the survey open-ended responses indicated that the respondents did not remember seeing a final report on the evaluation. One person wrote, "I was not involved in the planning or implementation of the program evaluation, and I don't believe I have ever seen the report of the evaluation." Similarly, another wrote, "I have never gotten/seen any info about CETP program evaluation findings."

The responses above highlight two issues that confronted the core evaluation. First, as described in the contextual material of this case study, the Core intended to host a third meeting in which the evaluation's final results would be shared. However, an accounting error occurred resulting in the Core project being informed that there were not enough funds to support further activities. The third meeting, consequently, was not scheduled, and work on the project was temporarily halted. Upon discovery of the error, work resumed on dissemination of the findings, including the writing of an article that was published in 2007[xv]. Although a full final report on the CETP program was produced, several of the project leaders and evaluators interviewed indicated that they did not remember seeing a copy.

High Users' Perspectives

This section of the case study presents data from interviews with survey respondents designated by the research team to be a "high user" based on having a mean use/influence score greater than 2.5 out of 4.0. Interviewees were selected based on mean use/influence score only, without consideration of involvement levels or role. In the end, three individuals – one project PI, one internal evaluator, and one external evaluator – consented and participated in a follow-up interview. The presentation of the high user interview data is organized around the issues that guided the data collection and analysis for the study and were discussed in detail in the sections above. They include:

- Motivation for Involvement
- Levels of Involvement
- Perceived Effect of Involvement
- Use by Project Leaders and Evaluators
- Use by Projects
- Factors Affecting Use

Motivation for Involvement. Both the internal evaluator and external evaluator had worked with Frances Lawrenz in her capacity as an external evaluator prior to the core evaluation being implemented. The project PI got involved when Frances provided training at an NSF sponsored PI meeting in Washington, DC. In all three cases, the interviewees stated that the core evaluation was appealing to them as they had already been doing evaluation activities with mixed success and appreciated the coordinated effort provided by the core evaluation. As the project PI stated, "the evaluation had been the thorn in our sides from the beginning because that's something we really thought a lot about and when we talked to other people it seemed like evaluation was being done in a tremendous number of different ways. There didn't seem to be any standardization. So I think the core evaluation was something that we were really all thankful for. It gave us some common ground" (CETP Interviewee 6). One of the evaluators also admitted that it was Frances herself who motivated him to get involved. Her national reputation and his previous experience working with her were compelling reasons to get involved with the core project.

Levels of Involvement. As described in this case study, the core evaluation protocols and procedures were developed as part of a collaborative process involving input from all interested projects. The three projects represented by the three high-use interviewees were all projects which participated in the national planning meetings during which the instruments were developed. One of these projects also conducted a field test of the instruments at their campus. Additionally, the interviewees reported using the instruments to gather data about their project and report it to the core for cross-project evaluation. One evaluator also reported providing feedback on initial results that were shared by the Core.

The three high-use interviewees ranged in their reported level of involvement from some to extensive. Each felt they were fairly highly involved, although two stated that they felt other

projects were even more highly involved or had more to contribute to developing the core evaluation instruments than they had.

Use by Project Leaders and Evaluators. All three interviewees felt that the core evaluation used their input in the development of the instruments and processes for the overarching effort. As the project PI wrote,

> I think it [the project's input] was used considerably. I know we had some really lively discussions at some point with people saying we can't get that data – we can't do that. We'd go back and forth about how we'd get things. The whole method of teacher observation was a discussion for a long time so I think a lot of that was kind of active discussion that kind of rounded things into shape. Frances has wonderful ideas and wonderful view of it, so it really was very different than our point of view and that part was very good also. (CETP Interviewee 6)

One of the PIs stated that

> everybody was able to give input and that there was a lot of effort made to incorporate most of the input that was generated except where there was a lot of redundancy. Then we all worked closely to work out the redundancies so that everything was represented but maybe it wasn't exactly what one person had submitted. (CETP Interviewee 7)

Use by Projects. In contrast to previously presented perspectives that the Core had little impact on the projects, two of the high-user interviewees reported that the core evaluation had an extensive amount of impact on their projects and the third reported it had between some and an extensive amount of impact. Each of the interviewees shared multiple specific examples of how their projects were affected by the Core.

The external evaluator shared that the instruments developed by the Core had a large impact on the evaluation he conducted of the project. He felt that the instruments were important measures that allowed for significant differences in outcomes to be discovered. He mentioned that he still frequently refers to the protocols that were developed and has modified them to be used in other projects as well.

The internal evaluator described that the core evaluation helped her to consolidate a previously fragmented evaluation effort. They had worked with a number of external evaluators during their early years as a project, but their participation in the Core, "made it much easier to target in on things that we should amend as we went along… [the Core] provided a variety of perspectives from various CETPs but used in a more seamless manner" (CETP Interviewee 7). Additionally, the project changed some of the original plans as a result of hearing findings from the core evaluation. They modified their logic model in order to increase the impact of their project. Most importantly, this project's participation in the Core made them more credible to their Board of Regents which helped them to get their proposals approved for additional project activities.

The project PI also cited a number of ways in which the Core affected his project. He shared that the core data allowed them to see new results that they had not previously seen. This allowed them to identify areas in which their teaching methods were successful, as well as courses that had "slipped a little bit" (CETP Interviewee #6). As a result, the project shifted its focus from science to mathematics. The PI reported that the core data allowed him to see how little they had changed mathematics teaching in comparison to science instruction. The data allowed his project to meaningfully discuss the differences between mathematics and science instruction and to agree as a project to move forward with focusing on math. Like the external evaluator, the project PI reported that the protocols provided a framework that has been used in another context within his university. In this case, the PI was on the university's promotion and tenure committee and introduced a time-interval faculty teaching observation protocol that was inspired by the CETP classroom observation protocol.

Factors Affecting Use. All three of the interviewees reported that the core evaluation had an extensive amount of impact on them personally. All three mentioned that they increased their evaluation skills and benefitted from the experience of working collaboratively with other CETP projects to develop the core evaluation instruments.

The external evaluator stated that the Core had a very strong effect on him and what he has done in his subsequent work. He said that he learned a lot through the process of working collaboratively to develop the core instruments and protocols with other projects, including building his evaluation skills and techniques. He also stated that he learned about leadership as a result of attending the meetings. He stated that Frances modeled for him how to get projects involved in evaluations and how to motivate individuals to participate meaningfully in evaluations.

The internal evaluator felt she benefitted most from the experience of working collaboratively with the other CETP projects by sharing data and ideas. She also stated that she "definitely" learned new evaluation skills and knowledge about STEM education. She learned about the innovative approaches other projects were implementing in more in-depth ways than just through yearly project meetings hosted by NSF.

The project PI, a theoretical scientist, stated that he was new to educational evaluation so his participation with the Core had a strong influence on his way of thinking. For example, his perspective on the value of evaluation changed. As he stated, "previously I thought of evaluation as something that had to be done so that the funding agencies could show the value of the program. I really hadn't used it as much to examine what I was doing myself. I think that was the big change (CETP Interviewee 6).

Project-Level Summary

This section of the case study summarizes the ways in which CETP project leaders and evaluators felt involved in the CETP program evaluation and subsequently used the results. One

major issue, collaboration, was identified as contributing to the core evaluation's effect on the projects. Four broad issues were identified as limiting the impact of the core evaluation on the projects.

Project leaders and evaluators were involved primarily in two stages of the core evaluation. First, project leaders and evaluators from mature CETPs were asked to become involved in the planning and instrument development phase of the evaluation. Second, all CETPs were asked to gather data and provide it to the Core. A majority of respondents reported being involved at least a little or some in all of the aspects of evaluation planning, implementation, and communication, although between 22% and 48% of survey respondents reported that they were not involved in any individual activity.

CETP project leaders and evaluators felt that their involvement in the program evaluation was the most important factor in terms of enabling the core evaluation to have an impact on them and their projects. Project leaders and evaluators involvement in the initial planning meetings, as well as their attendance at follow-up instrument training meetings, led them to feel committed to the evaluation process and to have confidence in the data collection instruments. The quality of the instruments and of the core evaluation team helped some project leaders and evaluators to have confidence in the instruments. Additionally, meeting attendance was felt to facilitate a broadening of perspectives on issues facing the field. Without engaging first-hand in the development of instruments to collect common data across disparate projects, some project leaders and evaluators were critical of the instruments and felt they were not representative of the outcomes produced by their projects.

The interviews and surveys of the project leaders and evaluators identified four major limitations to the CETP core evaluation having an effect on the projects. First, NSF's timing and implementation of the core evaluation as well as political issues that affected the evaluation's impact were mentioned. Second, the timing of the core evaluation in a project's funding cycle was related to whether there was an opportunity to use data from the Core. Third, perceived irrelevance of the Core data affected their usefulness to projects. Fourth, issues surrounding the dissemination of the Core's final findings limited the extent to which the projects could use the data.

Influence on the Field

This section of the case study examines factors that appear to be related to the use and influence of the CETP core evaluation on the fields of STEM education and evaluation. It presents findings resulting from four different data collection activities and is organized according to the following three topics:

- Projects' Perceptions of Influence on the Field
- The "Field's" Perception of Influence on the Field
- Evaluation Products' Influence on the Field

First, all survey respondents and interviewees were asked directly about their perceptions of the impact of the core evaluation on the field. Second, to assess the influence of these evaluations on the broader fields of STEM education and evaluation, the research team used two data collection methods: a survey of NSF PIs and a brief survey of relevant journal editors. The NSF PI survey was designed to assess whether the CETP evaluation, its PIs, and related publications would appear among the names and titles of influential evaluation-related people, products, and activities. The brief email/telephone survey of journal editors of six journals in the evaluation and STEM education fields was conducted to determine editor awareness of influential STEM evaluations. Finally, as another measure of influence on the field, citation analysis was conducted to examine the extent to which the CETP evaluation products were cited in scholarly articles in the field.

Projects' Perceptions of Influence on the Field

Overall, the three interviewees were in agreement that the CETP core evaluation had a large impact on themselves, their projects, and the fields of STEM education and evaluation. There appear to be two reasons that predominantly account for the evaluation's influence in the minds of these participants. First, all three of the interviewees stated that the tools and instruments produced by the evaluation were important for creating changes in their projects. One interviewee stated that tools were seen as "different than the standard measures" in that they were instruments that would "resound with faculty who saw merit in what was being measured, as opposed to a more standardized testing approach" (CETP Interviewee 8). Two interviewees commented that the development of a common measure to evaluate the projects, but one that permitted flexibility to tailor the evaluation to the local context, allowed the evaluation to be influential. Developing tools to capture nuanced changes in instruction had been a challenge for the projects and the core evaluation instruments were helpful in meeting those needs.

Second, the atmosphere of the meetings was also mentioned by all three interviewees as an important factor for the evaluation's influence. The meetings were characterized as being relaxed and collegial, but at the same time rigorous and academic. The participants felt appreciated for their contributions and because the tools that were developed arose out of the national evaluation effort, they had more cache locally than those generated by the individual sites.

The overall impact on all three levels was best put in the words of the internal evaluator. She explained that the CETP core evaluation

> …influenced me to work closely with other people who were doing evaluations in higher education, which was something I had not had a lot of experience with before that. Working as an independent evaluator was what I did but not with people working on all the same projects as I was and looking from it from different perspectives. That was very good. I consider myself lucky to have been able to participate in the core evaluation.

> In terms of the effect it had on our program, I think it was substantive and substantial because not only from the Board of Regents perspective on our standing which was increased; it was also at the level of the performances the disciplinary departments within the university support to cause better understanding. I think it enable me to grow within my position as an evaluator and to look at things with a broader view but it also enabled us to have more confidence to amend our program as we had to given the results that came in. (CETP Interviewee 7)

This case presents evidence of the impact of the core evaluation on individual project leaders' and evaluators' knowledge, skills, attitudes, and actions. First, this case study lends support to the assertion that involvement in the planning stages of a multi-site evaluation is related in important ways to whether the evaluation process influences individual project leaders and evaluators in meaningful and lasting ways. Leaders and evaluators who were involved in the evaluation planning process reported gaining knowledge and understanding of evaluation processes as well as gaining valuable evaluation skills. Additionally, the commitment to and belief in the value of the overall program evaluation appears to be enhanced by participation in the development of data collection instruments, while exclusion from this process may lead to some individuals questioning the quality and value of common data collected from multiple sites.

According to some individuals interviewed and surveyed for this case study, the core evaluation did not appear to have a very large impact on the individual CETP projects move or on the STEM education and evaluation fields, while for others the evaluation's influence was extensive. This variation can be explained to some extent by whether individual projects were at a point in their lifecycle to benefit from being part in the collaborative process of developing the core evaluation instruments and protocols.

There were several other issues involved in the initial conceptualization and implementation of a core program evaluation at the NSF level, as well as issues affecting the dissemination of evaluation findings by the Core that prevented the core evaluation from having a greater impact at the project level. First, the way in which NSF handled evaluation connected with the CETP program appears to have played a large role in the limited impact of the core evaluation on the projects. Second, timing issues played a role in whether the projects were able to use data from the Core. A third factor that appeared to affect the extent to which the core evaluation had an impact on a particular project was the extent to which project management believed that the CETP core data were applicable to their project context. Finally, the way in which the core findings were disseminated appeared to limit the amount of influence they had.

Project leaders and evaluators were asked to reflect on the impact that the core evaluation had on the STEM education field and evaluation community. It was difficult for project leaders and evaluators to speak to the impact on the evaluation and STEM education communities; however, as they felt that they did not have broad enough understandings of the fields to comment accurately. Additionally, as one interviewee explained, it is hard to pinpoint changes

in the field specifically to the CETP core evaluation. As she stated, "At some point it becomes unconscious, like stuff that we read. I'm in math education and a lot of stuff that we read, it's like yeah, yeah, yeah, we've been seeing that stuff for 12 years. So, you know, it doesn't jump out at you anymore." (CETP Interviewee 1)

One respondent, however, felt strongly that the core evaluation was instrumental in improving the culture of STEM teaching due to bringing educators together to talk about common issues as part of the evaluation planning and training activities. As the interviewee explained,

> Everybody was teaching in inefficient ways and [they] weren't concerned about teacher preparation as we all should be. And yet there was a core group of faculty in each university who each championed the cause. And I think that, more than anything, a result of the NSF project was that over the whole country there was a change in the culture of teaching. We can see it today. It was very influential. And I think that Frances was instrumental in tying all of us together. I would say more than the NSF staff actually (laughs). So she was kind of the outsider who came in, and her intent was to do the core evaluation, but I think the underlying thing that happened was that she got us communicating and thinking about the overall issue. I think there was a very… a usefulness to what she did. (CETP Interviewee 2)

As shown in Figures 1 and 2, survey responses to questions about the influence of the core evaluation on the broad STEM education and evaluation communities were similar, with the greatest proportions of respondents stating that it was marginally or somewhat influential. The mean rating of the core evaluation's influence on the STEM education community was 2.86, while the mean rating of the evaluation's influence on the evaluation community was slightly lower at 2.65.

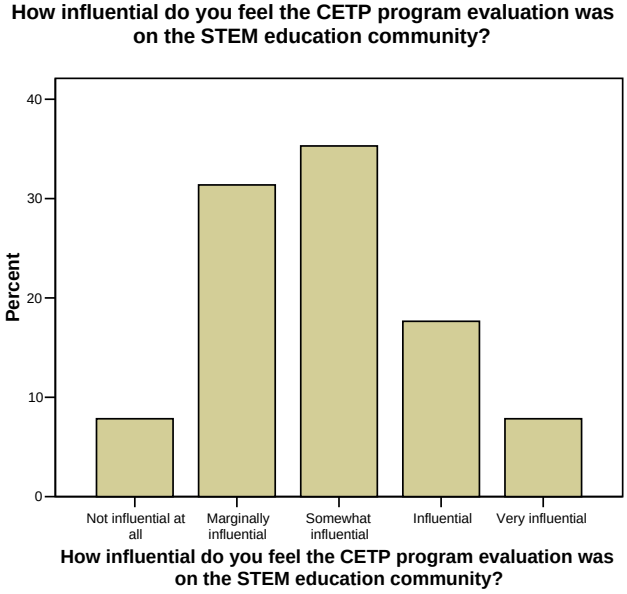


Figure 1. Influence of the core evaluation on the STEM education field

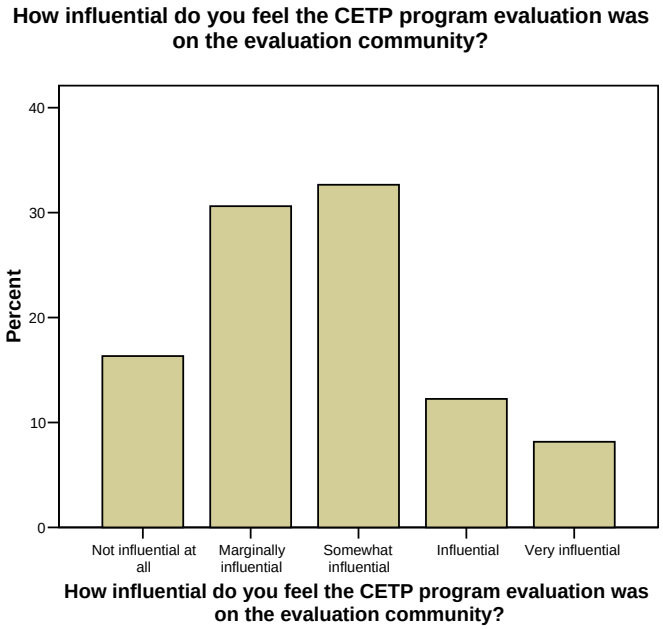


Figure 2. Influence of the core evaluation on the evaluation community

All three interviewees stated that the development of common, standardized, and quantitative indicators of progress in science and math education was an important impact on the STEM field.

One of the interviewees characterized the CETP core evaluation as a "new style" assessment program that provides an important contrast to the "non-authentic" style of educational evaluation represented by No Child Left Behind. Additionally, he believed that the CETP program was an important part of the larger movement of educational reform and there is a residual of the practices the program encouraged, affected to some extent by the core evaluation, that still exist today.

Both of the other two interviewees cited specific ways in which the core evaluation had an impact on policy makers within their states. In the first example, the project shared their classroom observation data with the state science and math coordinators who then passed the data on to the Secretary of Education. While he was uncertain about the exact outcome of the exchange, he stated that that is the kind of thing that "goes into shaping policy" (CETP Interviewee 6).

The second interviewee shared that her state's Board of Regents modeled an annual review of STEM participation in statewide colleges on the CETP core evaluation. The Board of Regents felt that the model was credible and rigorous because it was a nationally conducted evaluation effort supported by NSF. As she explained, the core evaluation served to

> broaden the importance of evaluation across proposals that were funded by the Board of Regents and then to force the colleges and the STEM proposals to know that they were going to be looked at every year. A cultural change had come about… and as a result, the level of awareness about evaluation within the state was raised. (CETP Interviewee 7)

The "Field's" Perception of Influence on the Field

The research team attempted to measure the field's perception by surveying NSF PIs and journal editors. The first of these two was the NSF PI survey which gathered PIs' perceptions related to evaluations, people and publications. The researchers reasoned that if respondents mentioned the names and titles of the CETP program evaluation, then this might indicate the presence of a network of influence among the four targeted programs and STEM educators and evaluators.

The research team examined the survey results for evidence of influence of the core CETP evaluation, its PIs (PIs), and related publications on the broader fields of STEM education and evaluation. Sixty-four of the 191 respondents indicated they were able to identify an evaluation by name, but only 51 of the 64 respondents actually provided the name of an evaluation in a follow-up question. Within these 51 named evaluations, the CETP evaluation was named once. In addition, researchers asked survey respondents to note STEM evaluation publications they viewed as influential. Only one of 44 respondents named a publication associated with CETP. Finally, the team asked respondents to identify influential STEM evaluators and three of 59 respondents named an evaluator associated with CETP.
Focusing further, there were four respondents who reported that they were involved in CETP, one respondent identified as influential a CETP-affiliated evaluation, publication, or evaluator by name.

As a second look at the field's perception of influence on the field, the researchers conducted a 3-question survey of STEM education and evaluation journals. One of the three editors who responded to the journal editor survey was able to name an influential evaluation of a large scale, multi-site STEM education evaluation and that evaluation was not the CETP program evaluation.

Evaluation Products' Influence on the Field

Our final measure of influence on the field considers the extent to which the CETP evaluation products were cited throughout the field. The citation analysis uses bibliographic methods to gather data about the use of program evaluation products by individuals within the STEM education and program evaluation communities. Citation analysis consists of tracking the number of citations other authors make to a published work.

This section describes the 24 products produced by the CETP evaluation and the 42 unique publications (referred to as "citing works") that cited one or more of the CETP evaluation

products. Following the product and citing work descriptions, patterns in the citation data are explored through network maps.

Descriptions of the CETP Core Evaluation Products. The CETP evaluation produced a total of 24 evaluation related products. A list of these products and the citing works to each product are presented in Appendices A and B. The products were coded as to their types, fields, and subfields according to the definitions presented below:

Evaluation products were coded by type according to the following:

- Instrument/tools: evaluation instruments (observation protocols, surveys, etc), instrument manuals, online evaluation tools (logic model tool, resource database);
- Presentations: conference presentations, presentations to NSF or other audiences;
- Publications: journal articles, books, book chapters, published monographs, dissertations, newsletters;
- Reports: evaluation reports, briefing papers, fact sheets, brochures.

Product fields were broadly defined according to the following:

- STEM education/research: general topics or research related to STEM education;
- STEM evaluation: topics or findings connected to a specific STEM evaluation;
- Education—general : education areas/topics outside of STEM education;
- Evaluation—general : evaluation areas/issues outside of STEM evaluation;
- Other: items not falling into any of the above categories

Products were coded as to the specific sub-field within education or evaluation according to the following:

- Professional development: teacher training during service including the effects of professional development on student achievement, etc.;
- Workforce development: high tech, STEM workforce training/development;
- Partnerships/systemic reform: university, school, community partnerships, or state or district-wide systemic reform initiatives;
- Curriculum/instruction/assessment: related to classroom teaching (i.e., instruction methods, curriculum, pedagogy, student assessment, standards-based instruction, math/science reform);
- Evaluation: evaluation-related topics such as multi-site evaluation theory or other general evaluation theory or knowledge, including evaluation instruments, protocols, and manuals;
- Materials development: development of instructional materials;
- Project capacity building: issues related to program improvement, sustainability, advisory committee development, dissemination, replication;
- Other: other fields.

**Please note that evaluation instruments, manuals, and protocols were coded as "evaluation" rather than by the topic the instrument collected data about.*

CETP Core Evaluation Product Types. The 24 products produced by the CETP core evaluation represented a variety of dissemination forms. The core evaluation research team presented a total of 10 presentations as part of the evaluation dissemination efforts, and five instruments, five reports, and five publications were produced (see Table 15).

Table 15. Types of products produced by the CETP core evaluation

Product Type	N	%
Presentation	10	41.7
Instrument/Tool	5	20.8
Report	5	20.8
Publication	4	16.7
Total	*24*	*100.0*

The CETP core evaluation primarily produced products related to the CETP evaluation efforts; twenty of the twenty-four products were evaluation reports, instruments, or other products presenting and sharing information about the core evaluation's methods and findings (see Table 16). Additionally, the CETP core published one article and presented one conference paper related to general evaluation issues. Both the article and presentation concerned issues in conducting national multi-site evaluations. The CETP core evaluation also published two articles in STEM education journals regarding STEM education issues in general. One was about the use of technology at different grade levels and the second discussed learning in introductory science classes.

Table 16. Fields of evaluation products

Product Fields	N	%
STEM evaluation	20	83.3
Evaluation - general	2	8.3
STEM education/research	2	8.3
Total	*24*	*100.0*

The products produced by the CETP core evaluation were rather narrowly focused, as opposed to addressing a wide range of STEM educational issues, as was evident in some of the other

evaluations under consideration in this study (see Table 17). Almost 80% of the products produced by the core evaluation were related to evaluation issues, methods, and instruments in some way. Additionally, three products concerned teacher training issues and two addressed issues within the areas of curriculum and instruction.

Table 17. Specific fields of evaluation products

Product Subfields	N	%
Evaluation	19	79.2
Teacher training/Pre-service	3	12.5
Curriculum/Instruction/Assessment	2	8.3
Total	*24*	*100.0*

Citations to the Evaluation Products. Ten of the twenty-four CETP core evaluation products received one or more citations. One of the products, the CETP Classroom observation protocol received 27 citations; while the remaining nine products received one to three citations each (see Table 18).

Table 18. Number of citations per product

Number of citations	Number of Products	%
0	14	58.3
1	5	20.8
2	2	8.3
3	2	8.3
27	1	4.2
Total	*24*	*100.0*

It is interesting to note that five of the ten cited evaluation products were related to evaluation instruments created by a collaborative process between the CETP core evaluation and a number of CETP projects, including the Classroom Observation Protocol, the Handbook for Evaluation of Activities and Assessments, the Higher Education Survey Handbook, the K-12 Surveys Handbook, and the Classroom Observation Protocol Video Instruction Package.

Significant differences were not found in the mean number of citations received by products of different types, fields, or subfields. On average, each CETP product received 1.75 citations.

That said, it is interesting to note that none of the CETP presentations received citations, so in reality each of the 14 instruments, publications, and reports averaged 3 citations each.

Description of the Works Citing the CETP Core Evaluation Products. A total of 42 references were found to the ten cited evaluation products. The 42 references came from 38 unique citing works, which are detailed in Appendix B. The vast majority of citing works cited only one core evaluation product (n=34). There were also four citing works that cited two evaluation products. All of four citing works were written by authors not affiliated with the CETP core evaluation team. These data suggest that the knowledge generated by the CETP core evaluation was rather diffused in nature, as opposed to being concentrated in terms of being highly cited within a smaller number of works.

As shown in Table 19, over 40% of the works citing products from the CETP core evaluation were authored by individuals affiliated with CETP projects across the country. In total, there were nine different CETP projects that cited CETP core evaluation products. Almost one third of the citing works were from individuals not affiliated with the CETP evaluation, CETP projects, or other NSF funded grant projects.

The remainder of the citing works was connected in some way to the CETP core evaluation, the other three evaluations under consideration in the Beyond Evaluation Use study, or NSF. Two of the citing works were other CETP core evaluation products. Six of the citing works were written by either PIs or staff members affiliated with one of the program evaluations. One was a product produced by the ATE evaluation and one was an NSF Committee of Visitors report.

Table 19. Authors of citing works

Citing Work Authors	N	%
CETP Project	16	42.1
Non-affiliated	12	31.6
PI	5	13.2
CETP Core Evaluation	2	5.3
ATE Evaluation	1	2.6
NSF	1	2.6
Staff	1	2.6
Total	38	100.0

The citing works were predominantly US in origin, although two of the citing works were from Canadian sources. Citing works were also categorized by type according to the same

categorization scheme used with the products, with the addition of codes for "electronic resources" which was used for Web pages, archived listserv messages, or similar online sources, and the category "other." As shown in Table 20, citing works were divided among reports (n=13), publications (n=12), and presentations (n=9), with four citing works coming from online resources. Many of the reports were evaluation reports written by PIs and evaluators from the CETP projects who were involved with the CETP core evaluation development and instrumentation processes.

Table 20. Citing work types

Types	N	%
Report	13	34.2
Publication	12	31.6
Presentation	9	23.7
Electronic Resource	4	10.5
Total	*38*	*100.0*

As shown in Table 21, the majority of the citing works were connected to STEM evaluations or addressed STEM evaluation issues (n=25). STEM educational and research oriented publications were also well represented with eight citing works writing about general educational topics with the STEM areas. Four of the citing works addressed general issues in evaluation, outside of the areas of STEM, including issues in multi-site evaluations, developing classroom observation protocols for evaluative purposes, and one was a review of an evaluation journal that included a CETP publication. The remaining one publication addressed the topic of developing observational tools to measure cooperative learning.

Table 21. Citing work fields

Fields	N	%
STEM evaluation	25	65.8
STEM education/research	8	21.1
Evaluation	4	10.5
Education - general	1	2.6
Total	*38*	*100.0*

The citing works represented a variety of specific fields within education and evaluation, see Table 22. Like the evaluation products themselves, there were a relatively small number of

content areas addressed by the citing works. The citing works most often related to evaluation topics, including evaluation instruments (n=14). Teacher training and pre-service issues, which was the focus of the CETP program, were addressed by twelve of the citing works, while issues in curriculum, instruction, and assessment were covered in nine citing works. There was also one NSF committee of visitors report, one NSF grant report, and one publication about student engagement that made reference to the CETP classroom observation protocol.

Table 22. Citing work sub-fields

Subfields	N	%
Evaluation	14	36.8
Teacher training/Pre-service	12	31.6
Curriculum/Instruction/Assessment	9	23.7
NSF	2	5.3
Other	1	2.6
Total	*38*	*100.0*

Citation Patterns of CETP Core Evaluation Products. While the data presented above convey one picture of the citations to CETP core evaluation products, this section presents a graphical picture the citation network. For the following images, the program NetDraw was used to create maps of the networks of CETP evaluation products and citations. In each of the maps, arrows point from the citing work to the evaluation product it is citing. The products and citing works are in the same position in each of the maps; only the colors change to highlight different characteristics of interest to help describe citation patterns.

Figure 1 presents the overall citation structure of CETP core evaluation products (blue squares) and citing works (red circles). The figure highlights the impact of one of the CETP products (CETP012) – the CETP Classroom Observation Protocol. As mentioned above, this one product accounts for 27 of the 42 total citations. The figure also shows that there are three citing works (i.e., 1205, 1139, and 1051) which cite both the classroom observation protocol and one other evaluation product.

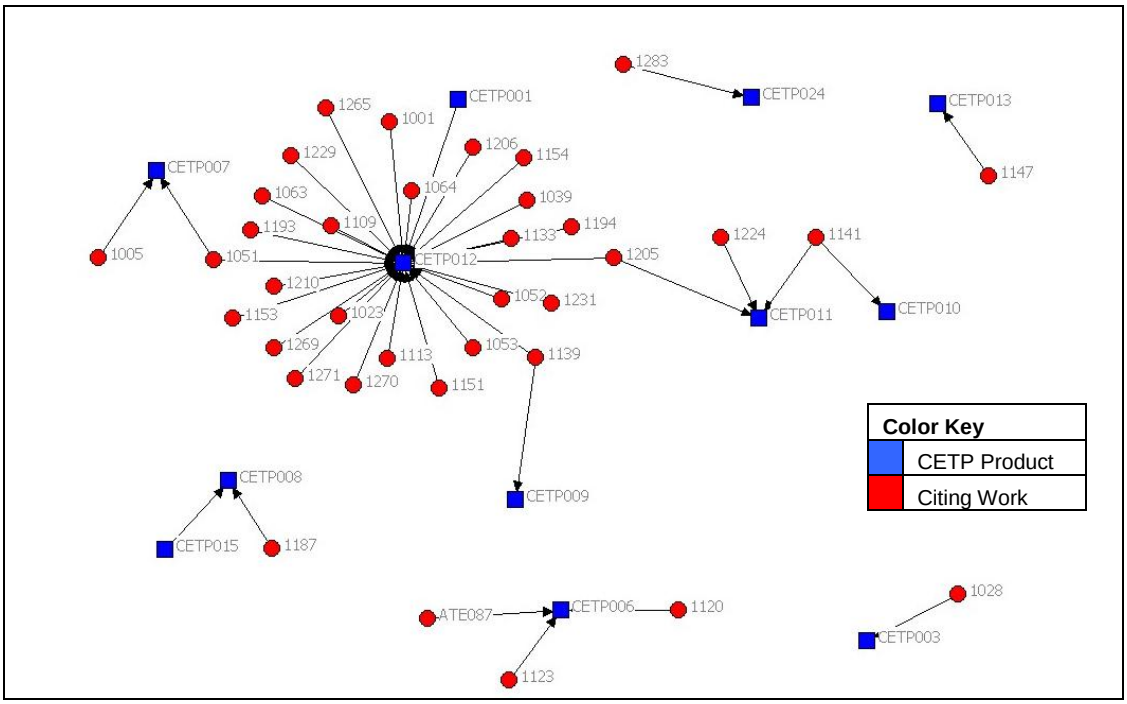


Figure 1. CETP core evaluation products and citations

In Figure 2, the color coding scheme is changed to illustrate the different authors of the external citing works. CETP Core evaluation products are again represented as blue squares, however other sources are colored according to the key. As illustrated in the figure, citations to the CETP Classroom Observation protocol were split between sources closely connected to the core evaluation (e.g., CETP projects or other NSF PIs or staff) and sources that were not affiliated with the core evaluation in any direct way.

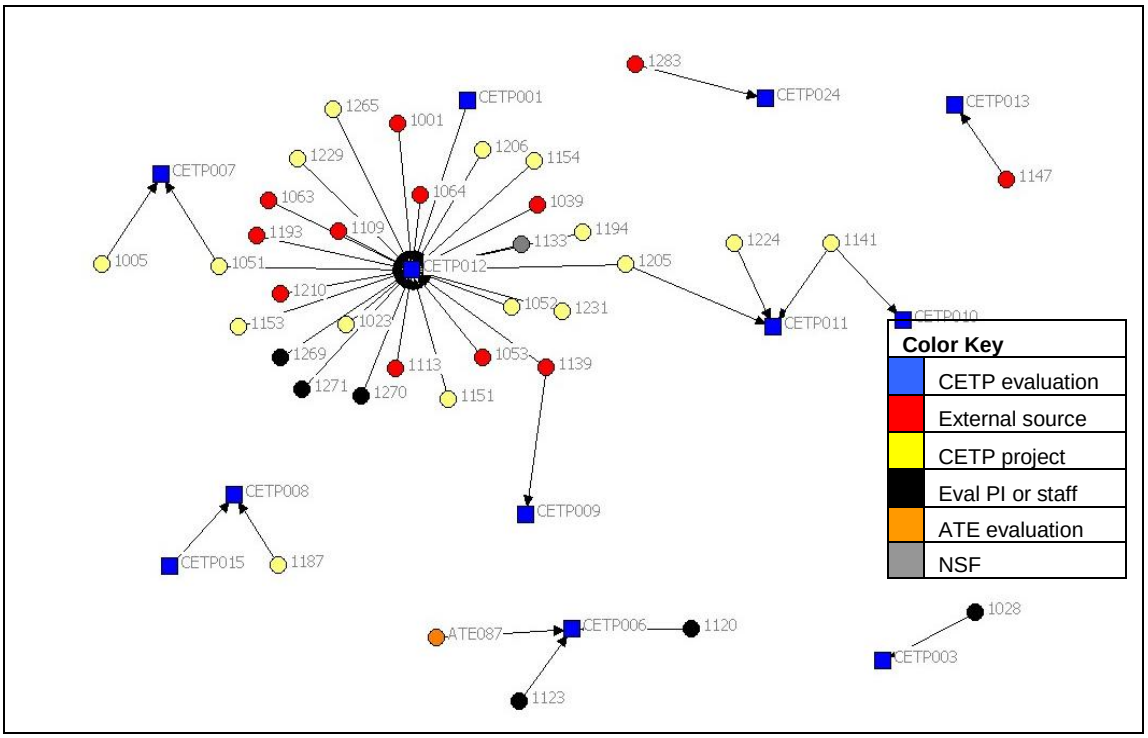


Figure 2. Citing authors of CETP core evaluation products

In Figure 3, the products and citations are all re-colored to represent the specific fields that the publication addressed; a color key is provided with the figure. The CETP core evaluation products are squares and other works are represented as circles.

As shown in the figure, many of the cited evaluation products were evaluation-related in nature, including several instruments. Many of the citing works were also related to evaluation issues. The fields of the works citing the CETP Classroom Observation protocol are somewhat varied, in that they address issues in all five of the field categories represented among the CETP products. The works citing the remaining CETP products, however, are concentrated in areas of evaluation and teacher pre-service. This suggests that while the Classroom Observation Protocol had interest in a number of areas, the rest of the CETP core evaluation products had somewhat limited appeal in terms of the fields that cited them.

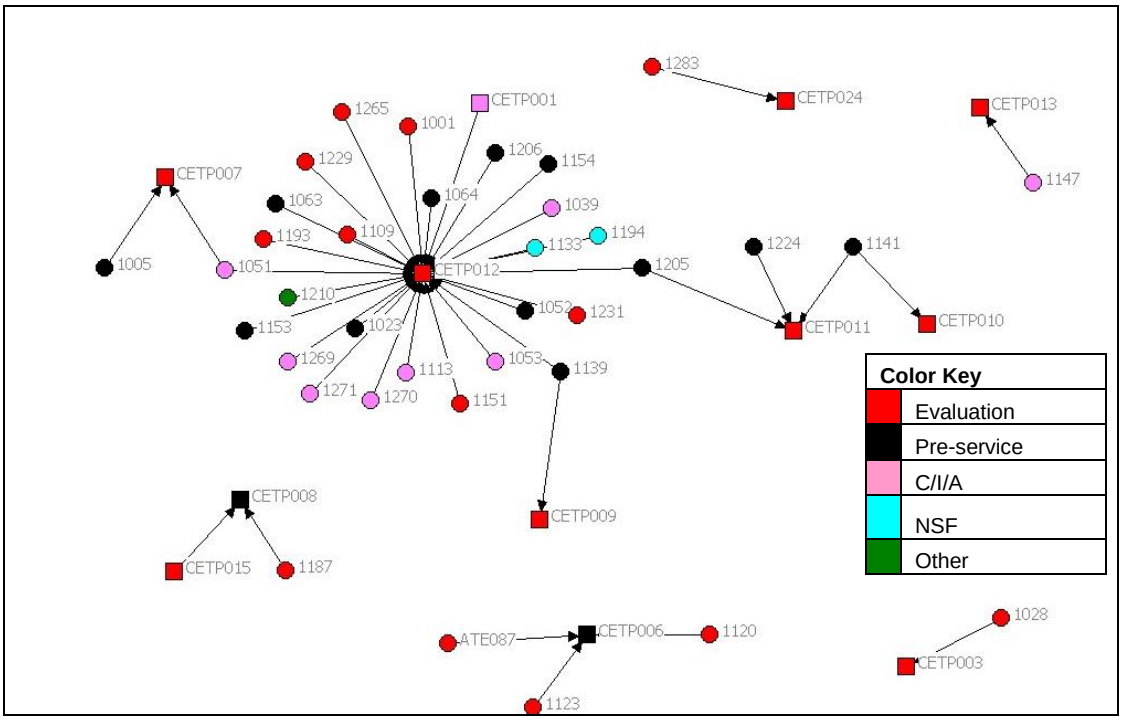


Figure 3. Fields of citing works and products

Citation Analysis Discussion and Conclusions. Dissemination was not a key priority for the CETP core evaluation, which was instead primarily concerned with creating an evaluation process and instruments collaboratively with the input of CETP project PI's and evaluators. That said, the evaluation instruments appear to have had some impact on the field of STEM education, at least within the area of pre-service teacher training. Among the CETP core evaluation products, the CETP Classroom Observation protocol was clearly most highly influential in terms of being cited within CETP projects as well as by unaffiliated sources. The remaining CETP products were picked up in one or two places but did not demonstrate such a high level of influence. In general, it appears that the evaluation instruments seemed to have the greatest influence on the STEM fields, particularly on individuals connected to the evaluation in some way.

Influence on the Field Summary

As supported by the data presented in the preceding sections, the CETP core evaluation appears to have had only a modest influence on the fields of STEM education and evaluation. Despite the fairly high level of project involvement, wide dissemination of results to the broader fields of STEM education and evaluation was not a key priority for the evaluation and funding issues held up final reporting efforts. The interviewed project personnel were mixed in their perceptions of effect with some reporting that the effect was small while others felt the evaluation's influence was extensive. The citation analyses showed that the 24 products produced by the evaluation had some influence in terms of being referenced or used. The instruments especially appear to have been influential.

The field study and the survey of journal editors revealed that evaluations in general were not viewed as influential. In keeping with that overall finding, few of the responding NSF PIs were familiar with the CETP evaluation processes or products although there was some evidence of recognition. Because of the small number of responding PIs that had been involved in the CETP core evaluation, it was not possible to determine if there were any relationship between influence and involvement although it is interesting to note that of the four current PIs who reported having been involved in CETP only one identified something from the CETP evaluation as influential.

Overall Summary

The national CETP evaluation was instituted after many of the individual CETP projects had been in operation for some time. Through meetings and other types of communication the CETP evaluation team attempted to involve the existing projects in the design of the evaluation, the development of evaluation questions, the construction of data collection instruments, and data collection and analyses. NSF was the primary 'client' for the evaluation and therefore the projects and the fields of STEM education and evaluation were secondary audiences not the directly intended user.

This case study showed a modest relationship between CETP project involvement in the CETP evaluation and influence of the evaluation's processes and products on the projects. It was clear that involvement was not the only variable affecting influence. Furthermore, the effect of involvement was confounded by a wide range of perceptions about what constitutes involvement. Despite similar opportunities to participate in CETP evaluation activities, interviewees from some projects felt that they were heavily involved while others felt they were less involved and the activities cited as contributing to the involvement varied in both breadth and depth. For some, in depth involvement in one activity was seen as constituting substantial involvement while for others participation in several different activities was necessary. It did appear from the surveys and interviews with project personnel, however, that perceptions of levels of involvement were positively linked to perceptions of use. Other variables affecting use included where the project was in its life cycle, the perceived relevance of the CETP evaluation to the local project and interactions with NSF.

The CETP evaluation had only a very small discernable effect on the fields of STEM education and evaluation. Although responses were mixed, overall through the eyes of the project personnel, the CETP evaluation was viewed as having a small effect on the fields of science education and evaluation and that this effect was probably diffused and not really attributable directly to the CETP evaluation. The citation analyses of use of the products of the CETP evaluation showed that there was some use particularly of the instruments by people in CETP projects and elsewhere. The survey of NSF PIs revealed that the respondents had little knowledge of evaluations in general and practically no knowledge of the CETP evaluation.

In summary, the CETP evaluation had a modest impact on the CETP projects that was enhanced by the involvement of the projects in the evaluation and only a very small effect on the fields of STEM education and evaluation.

Appendix A: CETP Core Evaluation Products

Product ID	Product Reference Information	Product Type	Product Field	Product Subfield	# Cites	Citation IDs
CETP001	Lawrenz, F., Gravely, A., and Ooms, A. (2006). Perceived helpfulness and amount of use of technology in science and mathematics classes at different grade levels. School Science and Mathematics, 106(3), 133-139.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
CETP002	Lawrenz, F., Huffman, D. and Appeldoorn, K. (2005). The context for learning in introductory science classes. Journal of College Science Teaching, 34(7), 40-44.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment	0	
CETP003	Lawrenz, F. & Huffman D. (2004). Using Multi-site Core Evaluation to Provide "Scientific" Evidence. Canadian Journal of Program Evaluation, 19(2),17-36.	Publication	Evaluation - general	Evaluation	1	1028
CETP004	Appeldoorn, K. (2004) Developing and validating the Collaboratives for Excellence in Teacher Preparation (CETP) core evaluation classroom observation protocol (COP). Doctoral Dissertation. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Publication	STEM evaluation	Evaluation	0	
CETP005	Lawrenz, F., Michlin, M., Appeldoorn, K., Goebel, A., & Schultz, M. (2004, March). CETP Core Evaluation: 2002-2003 Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Teacher training/Pre-service	0	
CETP006	Lawrenz, F., Michlin, M., & Appledoorn, K., (2003, March). CETP Core Evaluation: 2001-2002 Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Teacher training/Pre-service	3	ATE087, 1120, 1123
CETP007	Lawrenz, F., Huffman, D., Appeldoorn, K., & Michlin, M., (2002, February). CETP Core Evaluation: Field Test Results. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Report	STEM evaluation	Evaluation	2	1005, 1051
CETP008	Frantz, C., Lawrenz, F., Kushner, J., & Millar, S.B. (1999). Corridors for Collaboration. Washington, DC: National	Report	STEM evaluation	Teacher training/Pre-	2	CETP015, 1187

	Science Foundation.			service		
CETP009	Lawrenz, F., Huffman, D., and Appeldoorn, K. (2002). Classroom Observation Protocol Video Instruction Package. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1139
CETP010	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). K-12 Surveys Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1141
CETP011	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). Higher Education Survey Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	3	1141, 1205, 1224
CETP012	Lawrenz, F., Huffman, D., Appeldoorn, K., & Sun, T. (2001-2002). Classroom Observation Handbook. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	27	1001, 1023, 1039, 1051, 1052, 1053, 1063, 1064, 1109, 1113, CETP001, 1133, 1139, 1151, 1153, 1154, 1193, 1194, 1205, 1206, 1210, 1229, 1231, 1265, 1269, 1270, 1271
CETP013	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2002). Handbook for Evaluation of Activities and Assessments. Minneapolis, MN: University of Minnesota Center for Applied Research and Educational Improvement.	Instrument/ Tool	STEM evaluation	Evaluation	1	1147
CETP014	Lawrenz, F., & Appeldoorn. (2004, March). CETP Core Evaluation: Seeing a Glass Half Full. Presentation to the NSF Teacher Preparation PI Conference.	Presentation	STEM evaluation	Evaluation	0	
CETP015	Lawrenz, F. (2003, June). Differential Effects of Participatory Evaluation in a National Multi-site Program Evaluation. Presentation at the Canadian Evaluation Society national meeting, Vancouver, Canada.	Presentation	Evaluation - general	Evaluation	0	
CETP016	Lawrenz, F., Huffman, D., & Michlin, M. (2003, November). CETP Core Evaluation: Results Issues and the Juxtaposition of Scientific and Participatory Evaluation. Presentation at the American Evaluation	Presentation	STEM evaluation	Evaluation	0	

	Association National Meeting, Reno Nevada.					
CETP017	Lawrenz, F., & Huffman, D. (2002, November). Observing Standards-based Teaching in Science and Mathematics Classes in K-12 and Higher Education Settings. Presentation at the American Evaluation Association national meeting Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP018	Lawrenz, F. (2002, April). Multicultural Aspects in the Core Evaluation of the Collaboratives for Excellence in Teacher Preparation. Pathways to Change International Conference. National Science Foundation, CETP Program, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP019	Lawrenz, F., Huffman, D., & Appeldoorn, K. (2001, November). Core Evaluation of the Collaborative for Excellence in Teacher Preparation (CETP). Presentation at the American Evaluation Association National Meeting in St. Louis.	Presentation	STEM evaluation	Evaluation	0	
CETP020	Lawrenz, F., & Appeldoorn, K. (2001, April). Workshop on the Use of the Classroom Observation Protocol five hour training session. Presentation at the Collaboratives for Excellence in Teacher Preparation national PI meeting.	Presentation	STEM evaluation	Evaluation	0	
CETP021	Lawrenz, F. (2001, April). CETP Core Evaluation Project. Plenary session panel at NSF's Collaboratives for Excellence in Teacher Preparation national PI meeting, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP022	Lawrenz, F. (2000, March). CETP Core Evaluation. Presentation at NSF national meeting, Washington, DC.	Presentation	STEM evaluation	Evaluation	0	
CETP023	Lawrenz, F. (1998, November). Developing an evaluation core. Presentation at NSF sponsored workshop for Collaboratives for Excellence in Teacher Preparation evaluators.	Presentation	STEM evaluation	Evaluation	0	
CETP024	Lawrenz, F., Huffman, D., & Appeldorn, K. (2002). Evaluation of the Collaboratives for Excellence in Teacher Preparation (CETP). Minneapolis: University of Minnesota.	Report	STEM evaluation	Evaluation	1	1283

Appendix B: Citing Works to CETP Core Evaluation Products

Citation ID	Citating Work Reference	Citation Type	Citation Field	Citation Subfield
1001	633 Study Group. (nd). Chapter 2 Developing a Research Proposal. Available from: http://663studygroup.pbwiki.com/Chapter%202%20%20Developing%20a%20Research%20Proposal	Electronic Resource	STEM education/ research	Evaluation
1005	Adamson, S.L., Banks, D., Burtch, M., Cox, F., Judson, E., Turley, J.B., Benford, R., & Lawson, A.E. (2003, December). Reformed undergraduate instruction and its subsequent impact on secondary school teaching practice and student achievement. Journal of Research in Science Teaching, 40(10), 939-957.	Publication	STEM education/ research	Teacher training/ Pre-service
1023	Collaborative for Excellence in Teacher Preparation in Pennsylvania. (2002, June). CETP-PA Definitions & Terms: Definitions Related to Course Revision. Harrisburg, PA: author. Available from: http://www.cetp-pa.iup.edu/terms.htm	Report	STEM evaluation	Teacher training/ Pre-service
1028	Coryn, C.L.S. (2005, February). Global Review: Publications: Canadian Journal of Program Evaluation, Volume 19(2), Fall 2004. Journal of MultiDisciplinary Evaluation, 2.	Publication	Evaluation	Evaluation
1039	Dirr, P.J. (nd.) Classroom Observation Protocols: Potential Tools for Measuring the Impact of Technology in the Classroom. Policy and Planning Series #104. Alexandria, VA: Appalachian Technology in Education Consortium.	Report	Evaluation	Curriculum/ Instruction/ Assessment
1051	Flick, L., Morrell, P. D., & Wainwright, C. (2004). Reform Teaching Strategies Used by Student Teachers. School Science and Mathematics, 104(5), 199-213.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
1052	Flugman, B., Schroder, B., & Gningue, S. (2004, March.). Follow-up Study of the New York Collaborative for Excellence in Teacher Preparation. Presentation to the Teacher Preparation PI Conference, NSF CETP, STEMTP, ATE Programs, Arlington, VA.	Presentation	STEM evaluation	Teacher training/ Pre-service
1053	Fouts, J.T., Brown, C., & Thieman, G.Y. (2002, September). Classroom Instruction in Gates Grantee Schools: A Baseline Report. Bothell, WA: Fouts and Associates.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment

1063	Goss, M. (2003, May 1). Authentic Assessment of Pre-Service Students' Performance with a Palm or iPAC Handheld Tool. TechLearning Newsletter. Available from: http://www.techlearning.com/db_area/archives/WCE/archives/marlgoss.php	Publication	STEM evaluation	Teacher training/ Pre-service
1064	Goss, M., & Newcombe, E. (nd). Summary: NSF CETP-PA Level II Supervision Grant: Partnership for Excellence in Supervision. Available from: http://www.wcupa.edu/_ACADEMICS/CoEd/EdTech/cept_pa.htm	Report	STEM evaluation	Teacher training/ Pre-service
1109	Kern, A.L, Moore, T.J, & Akillioglu, F.C. (2007, October 10-13). Cooperative Learning: Developing an Observation Instrument for Student Interactions. 37th ASEE/IEEE Frontiers in Education Conference, Milwaukee, WI.	Presentation	Education - general	Evaluation
1113	Knapp, A.K. (2006). Activities in algebra: Is GK-12 effective for high school algebra students? Presentation made at the regional meeting of the National Council of Teachers of Mathematics, Chicago, IL.	Presentation	STEM education/ research	Curriculum/ Instruction/ Assessment
1120	Lawrenz, F. (2007). Review of science education program evaluation. In S. Abell & N.G. Lederman (Eds.), Handbook of Research on Science Education, (pp. 943-954). Mahwah, NJ: Lawrence Erlbaum.	Publication	STEM evaluation	Evaluation
1123	Lawrenz, F., & Huffman, D. (2006). Methodological Pluralism: The Gold Standard of STEM Evaluation. Critical issues in STEM Evaluation. New Directions for Evaluation, 109, 19-34.	Publication	STEM evaluation	Evaluation
1133	Lightbourne, J. (2004, June 15). Memorandum: COV for Teacher Preparation (CETP and STEMTP) Program: COI and Diversity Memo. Washington, DC: Directorate for Education and Human Resources, National Science Foundation. Available from: http://www.nsf.gov/od/oia/activities/cov/ehr/2004/DUE_TEcov.pdf	Report	STEM evaluation	NSF

1139	MacIsaac, D., Falconer, K., & Zawicki, J. (2005). Using the Reformed Teaching Observation Protocol (RTOP) to facilitate novice teachers in sustaining extraordinary levels of student discourse in introductory physics. Presentation at the American Association of Physics Teachers, Salt Lake City, Utah.	Presentation	STEM evaluation	Teacher training/ Pre-service
1141	Madfes, T.J. & Rosen, R. (2004, November). Follow-up evaluation of MASTEP [Mathematics and Science Teacher Education Program]: Final Report. San Francisco: WestEd.	Report	STEM evaluation	Teacher training/ Pre-service
1147	McClintock, E., O'Brien, G., & Jiang, Z. (2005, Summer). Assessing Teaching Practices of Secondary Mathematics Student Teachers: An Exploratory Cross Case Analysis of Voluntary Field Experiences. Teacher Education Quarterly.	Publication	STEM evaluation	Curriculum/ Instruction/ Assessment
1151	Miller, C., Stewart, M., & Townsend, J. (2004, March 14-15). Operational Approaches to the Observation Process. Presentation at the Teacher Preparation PI Conference: NSF CETP, STEMTP, ATE Programs.	Presentation	STEM evaluation	Evaluation
1153	Minnick, K.F., & McCormick, R. (2005, April 19). DRAFT TxCETP Evaluation Report 2000 - 2004. Albuquerque, NM: Minnick & Associates.	Report	STEM evaluation	Teacher training/ Pre-service
1154	Morrell, P.D., Flick, L., & Wainwright, C. (2004, May). Reform Teaching Strategies Used by Student Teachers. School Science and Mathematics, 104(5), 199-213.	Publication	STEM education/ research	Teacher training/ Pre-service
1187	Sawada, D., et. al. (2002, October). Measuring Reform Practices in Science and Mathematics Classrooms: The Reformed Teaching Observation Protocol. School Science and Mathematics, 102(6), 245-253.	Publication	STEM evaluation	Evaluation
1193	Science Education Partnership Award. (2006). SEPA-Generated Evaluation Tools from: Brain To Middle Schools. Available from: http://www.ncrrsepa.org/Eval/G_Eval.asp?GrantID=R25RR017315	Electronic Resource	STEM evaluation	Evaluation
1194	Science Storm. (nd). National Science Foundation Award #0119078: A Follow-up Summative Evaluation of the New York City Collaborative for Excellence in Teacher Preparation. Available from: http://www.sciencestorm.com/award/0119078.html	Electronic Resource	STEM evaluation	NSF

1205	Sireci, S.G., Zanetti, M.L., Slater, S.C., & Berger, J.B. (2001, September). STEMTEC Evaluation Report For Year 4 (Fall 2000/Spring 2001). Amherst, MA: University of Massachusetts Amherst.	Report	STEM evaluation	Teacher training/ Pre-service
1206	Slater, S.C., Sireci, S.G., Berger, J., Brown, M., O'Meara, K., Hintze, J., & DeCecco, L. (2003, August). STEMTEC II Evaluation Report for Year 1 -- Fall 2002/Spring 2003. Amherst, MA: University of Massachusetts Amherst.	Report	STEM evaluation	Teacher training/ Pre-service
1210	Soni, G. (nd). Student Motivation, Attitude, Engagement or Efficacy. Raleigh, NC: Learning in a Technology-Rich Environment, North Carolina State University.	Electronic Resource	STEM education/ research	Other
1224	Tittle, C.K., Flugman, B., Pape, S.M, & Saxman, L. (2001, July). Internal evaluation of the New York Collaborative for Excellence in Teacher Preparation: Final Report. New York: City University of New York, Center for Advanced Study of Education.	Report	STEM evaluation	Teacher training/ Pre-service
1229	Wainwright, C. (2007, January 10). Development of a Standards-Based Classroom Observation Instrument. Presentation to the American Association of Physics Teachers conference.	Presentation	STEM evaluation	Evaluation
1231	Wainwright, C., Flick, L., Morrell, P.D., & Schepige, A. (2004). Observation of Reform Teaching in Undergraduate Level Mathematics and Science Courses. School Science and Mathematics, 104(7), 322-336.	Publication	STEM education/ research	Evaluation
1265	Bravo, M. (2004, March). PR-CETP: Evaluation in the Context of Controversies in the Field of Program Evaluation. Presentation to the Teacher Preparation PI Conference: NSF CETP, STEMTP, ATE Programs. Available from: k12s.phast.umass.edu/stemtec/piconf/Proceedings/Papers/Bravo.doc	Presentation	STEM evaluation	Evaluation
1269	Callow-Heusser, C., Allred, D., Robertson, D., & Sanborn, W. (2005, September 5). Evidence-Based Small-Scale Study Final Report: McDougal Littell Middle School Math Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1270	Callow-Heusser, C., Allred, D., Robertson, D., Borman, G., & Dowling. (2005, July 31). McDougal Littell Evidence-Based Small-Scale Study Final Report: 7th Grade Life Science Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment

1271	Callow-Heusser, C., Allred, D., Sanborn, W., & Robertson, D. (2005, September 6). Evidence-Based Small-Scale Study Final Report: McDougal Littell Algebra 1, 2nd Edition Technical Report.	Report	STEM evaluation	Curriculum/ Instruction/ Assessment
1283	Zubrowski, B. (2007, December). An Observational and Planning Tool for Professional Development in Science Education. Journal of Science Teacher Education, 18(6), 861-884.	Publication	STEM evaluation	Evaluation
ATE087	Hanssen, C. E., & Coryn, C. L. S. (2004, November). Theory-driven evaluations of multi-site programs. Paper presented at the meeting of the American Evaluation Association, Atlanta, GA.	Presentation	Evaluation	Evaluation
CETP001	Lawrenz, F., Gravely, A., and Ooms, A. (2006). Perceived helpfulness and amount of use of technology in science and mathematics classes at different grade levels. School Science and Mathematics, 106(3), 133-139.	Publication	STEM education/ research	Curriculum/ Instruction/ Assessment
CETP015	Lawrenz, F. (2003, June). Differential Effects of Participatory Evaluation in a National Multi-site Program Evaluation. Presentation at the Canadian Evaluation Society national meeting, Vancouver, Canada.	Presentation	Evaluation	Evaluation

Endnotes

[i] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[ii] Project "leaders" refers to project primary investigators, co-primary investigators, coordinators, directors, administrators, or project consultants.

[iii] Number of respondents who provided a response directly related to the question or program evaluation. Some responses referred to the survey itself or indicated that they did not have a comment by writing "none," or "no comment;" these types of responses were not considered direct and are not reported in this table.

[iv] Lawrenz, F., Huffman, D. & Graves, A. (in press). Impact of the Collaboratives for Excellence in Teacher Preparation Program. *Journal of Research in Science Teaching.* 2007, 44(9):1348-1369

[v] National Science Foundation. (1999). Teacher Preparation Awards: NSF Collaboratives for Excellence in Teacher Preparation Awards. Retrieved from http://www.nsf.gov/pubs/1999/nsf9996/start.htm#_Toc442862272

[vi] Lawrenz, F., Appeldoorn, K., Michlin, M. (2003, July). Progress Report: CETP Core Evaluation Project. Report to NSF. Made available by CETP Principal Investigator.

[vii] Lawrenz, F. (2003, June). Differential Effects of Participatory Evaluation in a National Multi-Site Program Evaluation. Paper presented at the 2003 Meeting of the Canadian Evaluation Society Conference, Vancouver, BC. Retrieved from http://www.evaluationcanada.ca/distribution/20030601_lawrenz_frances.pdf

[viii] Lawrenz, F. (2003, June). Differential Effects.

[ix] Lawrenz, F., Huffman, D., & Appeldoorn, K. (2002, July). Progress Report: CETP Core Evaluation Project. Report to NSF. Made available by CETP Principal Investigator.

[x] F. Lawrenz (personal communication EUG "site visit" interview). May 2, 2005.

[xi] Lawrenz, F., Appeldoorn, K., Michlin, M. (2003, July). Progress Report.

[xii] Lawrenz, F. (2003, June). Differential Effects.

[xiii] F. Lawrenz (personal communication "PI Twenty Minutes"). September 27, 2005.

[xiv] All means were calculated by assigning scores as follows: 1=No, 2=Yes, a little, 3=Yes, some, 4=Yes, extensively

[xv] Lawrenz, F., Huffman, D. & Gravely, A. (2007). Impact of the Collaboratives for Excellence in Teacher Preparation Program. *Journal of Research in Science Teaching*.